# asset-class-performance-fed-policy-cycles

February 24, 2026

## 0.1 Introduction

In this post, we will look into the Fed Funds cycles and evaluate asset class performance during tightening and easing of monetary policy.

## 0.2 Python Imports

[1]:
```
# Standard Library
import datetime
import os
import sys
import warnings

from datetime import datetime
from pathlib import Path

# Data Handling
import numpy as np
import pandas as pd

# Data Sources
import pandas_datareader.data as web

# Statistical Analysis
import statsmodels.api as sm

# Machine Learning
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

# Suppress warnings
warnings.filterwarnings("ignore")
```

## 0.3 Add Directories To Path

[2]:
```
# Add the source subdirectory to the system path to allow import config from␣
↪settings.py
current_directory = Path(os.getcwd())
```

```
website_base_directory = current_directory.parent.parent.parent
src_directory = website_base_directory / "src"
sys.path.append(str(src_directory)) if str(src_directory) not in sys.path else␣
↪None

# Import settings.py
from settings import config

# Add configured directories from config to path
SOURCE_DIR = config("SOURCE_DIR")
sys.path.append(str(Path(SOURCE_DIR))) if str(Path(SOURCE_DIR)) not in sys.path␣
↪else None

# Add other configured directories
BASE_DIR = config("BASE_DIR")
CONTENT_DIR = config("CONTENT_DIR")
POSTS_DIR = config("POSTS_DIR")
PAGES_DIR = config("PAGES_DIR")
PUBLIC_DIR = config("PUBLIC_DIR")
SOURCE_DIR = config("SOURCE_DIR")
DATA_DIR = config("DATA_DIR")
DATA_MANUAL_DIR = config("DATA_MANUAL_DIR")
```

## 0.4 Python Functions

Here are the functions needed for this project:

- bb_clean_data: Takes an Excel export from Bloomberg, removes the miscellaneous headings/rows, and returns a DataFrame.
- calc_fed_cycle_asset_performance: Calculates metrics for an asset based on a specified Fed tightening/loosening cycle.
- load_data: Load data from a CSV, Excel, or Pickle file into a pandas DataFrame.
- pandas_set_decimal_places: Set the number of decimal places displayed for floating-point numbers in pandas.
- plot_bar_returns_ffr_change: Plot the bar chart of the cumulative or annualized returns for the asset class along with the change in the Fed Funds Rate.
- plot_scatter_regression_ffr_vs_returns: Plot the scatter plot and regression of the annualized return for the asset class along with the annualized change in the Fed Funds Rate.
- plot_timeseries: Plot the timeseries data from a DataFrame for a specified date range and columns.
- summary_stats: Generate summary statistics for a series of returns.

[3]:
```
from bb_clean_data import bb_clean_data
from calc_fed_cycle_asset_performance import calc_fed_cycle_asset_performance
from load_data import load_data
from pandas_set_decimal_places import pandas_set_decimal_places
from plot_bar_returns_ffr_change import plot_bar_returns_ffr_change
```

```
from plot_scatter_regression_ffr_vs_returns import␣
↪plot_scatter_regression_ffr_vs_returns
from plot_timeseries import plot_timeseries
from summary_stats import summary_stats
```

## 0.5 Data Overview

### 0.5.1 Acquire & Plot Fed Funds Data

First, let's get the data for the Fed Funds target rate (FFR). This data is found in 3 different datasets from FRED.

[4]:
```
# Set decimal places
pandas_set_decimal_places(5)

# Pull Federal Funds Target Rate (DISCONTINUED) (DFEDTAR)
fedfunds_target_old = web.DataReader("DFEDTAR", "fred", start="1900-01-01",␣
↪end=datetime.today())

# Pull Federal Funds Target Range - Upper Limit (DFEDTARU)
fedfunds_target_new_upper = web.DataReader("DFEDTARU", "fred",␣
↪start="1900-01-01", end=datetime.today())

# Pull Federal Funds Target Range - Lower Limit (DFEDTARL)
fedfunds_target_new_lower = web.DataReader("DFEDTARL", "fred",␣
↪start="1900-01-01", end=datetime.today())

# Merge the datasets together
fedfunds_combined = pd.concat([fedfunds_target_old, fedfunds_target_new_upper,␣
↪fedfunds_target_new_lower], axis=1)

# Divide all values by 100 to get decimal rates
for col in fedfunds_combined.columns:
    fedfunds_combined[col] = fedfunds_combined[col] / 100

# Resample to month-end (as if we know the rate at the end of the month)
fedfunds_monthly = fedfunds_combined.resample("ME").last()

display(fedfunds_monthly)
```

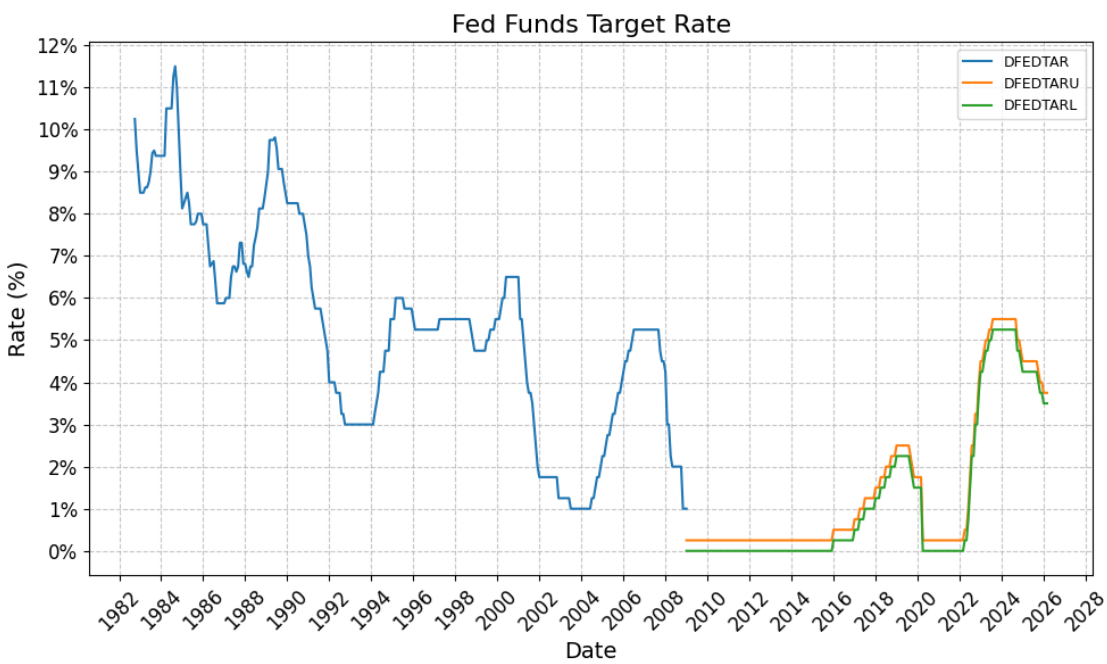
```
            DFEDTAR  DFEDTARU  DFEDTARL
DATE
1982-09-30  0.10250       NaN       NaN
1982-10-31  0.09500       NaN       NaN
1982-11-30  0.09000       NaN       NaN
1982-12-31  0.08500       NaN       NaN
1983-01-31  0.08500       NaN       NaN
…              …         …         …
```

```
2025-10-31       NaN   0.04000   0.03750
2025-11-30       NaN   0.04000   0.03750
2025-12-31       NaN   0.03750   0.03500
2026-01-31       NaN   0.03750   0.03500
2026-02-28       NaN   0.03750   0.03500

[522 rows x 3 columns]
```

We can then generate several useful plots. First, the Fed Funds target rate:

[5]:
```
plot_timeseries(
    price_df=fedfunds_monthly,
    plot_start_date=None,
    plot_end_date=None,
    plot_columns=["DFEDTAR", "DFEDTARU", "DFEDTARL"],
    title="Fed Funds Target Rate",
    x_label="Date",
    x_format="Year",
    x_tick_spacing=2,
    x_tick_rotation=45,
    y_label="Rate (%)",
    y_format="Percentage",
    y_format_decimal_places=0,
    y_tick_spacing=0.01,
    grid=True,
    legend=True,
    export_plot=False,
    plot_file_name=None,
)
```



[6]:
```
# First drop the lower column and merge
fedfunds_monthly["fed_funds"] = fedfunds_monthly["DFEDTAR"].
 ↪combine_first(fedfunds_monthly["DFEDTARU"])
fedfunds_monthly = fedfunds_monthly.drop(columns=["DFEDTARL", "DFEDTARU",␣
 ↪"DFEDTAR"])

# Compute change in rate from either column
fedfunds_monthly["fed_funds_change"] = fedfunds_monthly["fed_funds"].diff()

display(fedfunds_monthly)
```

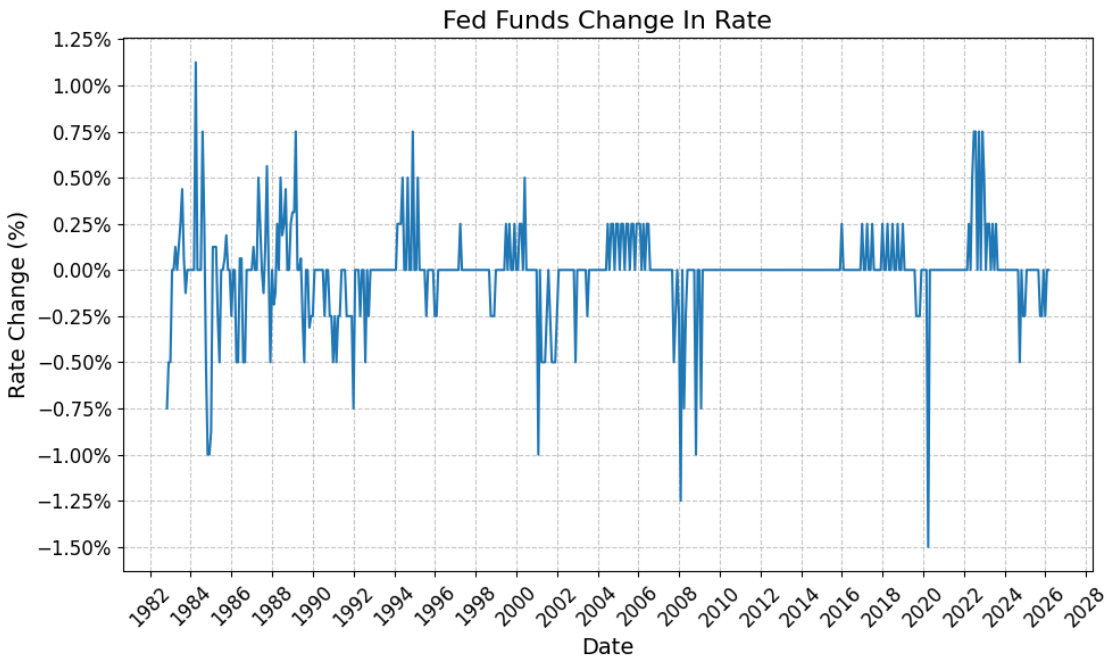
```
            fed_funds  fed_funds_change
DATE
1982-09-30    0.10250               NaN
1982-10-31    0.09500          -0.00750
1982-11-30    0.09000          -0.00500
1982-12-31    0.08500          -0.00500
1983-01-31    0.08500           0.00000
…               …               …
2025-10-31    0.04000          -0.00250
2025-11-30    0.04000           0.00000
2025-12-31    0.03750          -0.00250
2026-01-31    0.03750           0.00000
2026-02-28    0.03750           0.00000

[522 rows x 2 columns]
```

And then the change in FFR from month-to-month:

[7]:
```
plot_timeseries(
    price_df=fedfunds_monthly,
    plot_start_date=None,
    plot_end_date=None,
    plot_columns=["fed_funds_change"],
    title="Fed Funds Change In Rate",
    x_label="Date",
    x_format="Year",
    x_tick_spacing=2,
    x_tick_rotation=45,
    y_label="Rate Change (%)",
    y_format="Percentage",
    y_format_decimal_places=2,
    y_tick_spacing=0.0025,
    grid=True,
    legend=False,
```

```
    export_plot=False,
    plot_file_name=None,
)
```




This plot, in particular, makes it easy to show the monthly increase and decrease in the FFR, as well as the magnitude of the change (i.e. slow, drawn-out increases or decreases or abrupt large increases or decreases).

### 0.5.2 Define Fed Policy Cycles

Next, we will define the Fed policy tightening and easing cycles:

[8]:
```
# Set timeframe
start_date = "1989-12-15"
end_date = "2026-01-31"

# Copy DataFrame to avoid modifying original
fedfunds_cycles = fedfunds_monthly.copy()
fedfunds_cycles = fedfunds_cycles.loc[start_date:end_date]

# Reset index
fedfunds_cycles = fedfunds_cycles.reset_index()

# Drop the "fed_funds_change" column as we will use it to determine the cycle␣
↪type later, but we don't need it in the final dataframe
fedfunds_cycles = fedfunds_cycles.drop(columns=["fed_funds_change"])
```

```python
# Rename the date column to "Start Date", and the "fed_funds" column to "Fed␣
↪Funds Start"
fedfunds_cycles = fedfunds_cycles.rename(columns={"DATE": "start_date",␣
↪"fed_funds": "fed_funds_start"})

# Copy the "Start Date" column to a new column called "End Date" and shift it␣
↪up by one row
fedfunds_cycles["end_date"] = fedfunds_cycles["start_date"].shift(-1)

# Copy the "Fed Funds Start" column to a new column called "Fed Funds End" and␣
↪shift it up by one row
fedfunds_cycles["fed_funds_end"] = fedfunds_cycles["fed_funds_start"].shift(-1)

# Calculate the change in Fed Funds rate for each cycle
fedfunds_cycles["fed_funds_change"] = fedfunds_cycles["fed_funds_end"] -␣
↪fedfunds_cycles["fed_funds_start"]

# Based on "Fed Funds Change" column, determine if the rate is increasing,␣
↪decreasing, or unchanged
fedfunds_cycles["cycle"] = fedfunds_cycles["fed_funds_change"].apply(
    lambda x: "Tightening" if x > 0 else ("Easing" if x < 0 else "Neutral"))

display(fedfunds_cycles)
```

```
    start_date  fed_funds_start   end_date  fed_funds_end  fed_funds_change  \
0   1989-12-31          0.08250 1990-01-31        0.08250           0.00000
1   1990-01-31          0.08250 1990-02-28        0.08250           0.00000
2   1990-02-28          0.08250 1990-03-31        0.08250           0.00000
3   1990-03-31          0.08250 1990-04-30        0.08250           0.00000
4   1990-04-30          0.08250 1990-05-31        0.08250           0.00000
..         …                …        …             …                 …
429 2025-09-30          0.04250 2025-10-31        0.04000          -0.00250
430 2025-10-31          0.04000 2025-11-30        0.04000           0.00000
431 2025-11-30          0.04000 2025-12-31        0.03750          -0.00250
432 2025-12-31          0.03750 2026-01-31        0.03750           0.00000
433 2026-01-31          0.03750        NaT            NaN               NaN

       cycle
0    Neutral
1    Neutral
2    Neutral
3    Neutral
4    Neutral
..       …
429   Easing
430  Neutral
```

```
431   Easing
432  Neutral
433  Neutral

[434 rows x 6 columns]
```

Grouping the consecutive months of tightening and easing together, we can create a DataFrame that gives us the cumulative change in the FFR for each cycle, as well as the start and end dates for each cycle. This will be useful for evaluating asset class performance during these cycles.

[9]:
```
fedfunds_grouped_cycles = fedfunds_cycles.copy()

# Create a group key that increments whenever the Cycle value changes
fedfunds_grouped_cycles['group'] = (fedfunds_grouped_cycles['cycle'] !=␣
 ↪fedfunds_grouped_cycles['cycle'].shift(1)).cumsum()

# Group by both the group key and Cycle label
cycle_ranges = (
    fedfunds_grouped_cycles.groupby(['group', 'cycle'], sort=False)
    .agg(
        start_date=('start_date', 'first'),
        end_date=('end_date', 'last'),
        fed_funds_start=('fed_funds_start', 'first'),
        fed_funds_end=('fed_funds_end', 'last')
    )
    .reset_index(drop=False)
    .drop(columns='group')
)

display(cycle_ranges)
```

```
       cycle start_date   end_date  fed_funds_start  fed_funds_end
0    Neutral 1989-12-31 1990-06-30          0.08250        0.08250
1     Easing 1990-06-30 1990-07-31          0.08250        0.08000
2    Neutral 1990-07-31 1990-09-30          0.08000        0.08000
3     Easing 1990-09-30 1991-04-30          0.08000        0.05750
4    Neutral 1991-04-30 1991-07-31          0.05750        0.05750
..       …        …        …               …              …
116  Neutral 2024-12-31 2025-08-31          0.04500        0.04500
117   Easing 2025-08-31 2025-10-31          0.04500        0.04000
118  Neutral 2025-10-31 2025-11-30          0.04000        0.04000
119   Easing 2025-11-30 2025-12-31          0.04000        0.03750
120  Neutral 2025-12-31 2026-01-31          0.03750        0.03750

[121 rows x 5 columns]
```

Furthermore, we will make the assumption that any "Neutral" months (i.e. months where the FFR did not change) are part of the preceding cycle – to a point. For example, if we have a tightening month followed by a neutral month, we will consider the neutral month to be part of the

tightening cycle. This is a simplifying assumption, but it allows us to categorize all months into either tightening or easing cycles without having to create a separate category for neutral months. From a practical standpoint, this also makes it easier to evaluate asset class performance during these cycles, as we can simply look at the performance during the tightening and easing cycles without having to worry about the neutral months. We will foward fill no more than 6 months of neutral months, however, as it would be unreasonable to assume that a month of neutral policy is part of the preceding cycle if it has been 6 months since the last change in policy.

[10]:
```
fedfunds_grouped_cycles = fedfunds_cycles.copy()

# Forward fill cycle labels preceding "Neutral" cycles
fedfunds_grouped_cycles['cycle_filled'] = fedfunds_grouped_cycles['cycle'].
 ↪replace('Neutral', pd.NA).ffill(limit=6)

# Replaced any remaining <NA> values with "Modified Tightening"
fedfunds_grouped_cycles['cycle_filled'] =␣
 ↪fedfunds_grouped_cycles['cycle_filled'].fillna('Modified Tightening')

# Create a group key that increments whenever the Cycle value changes
fedfunds_grouped_cycles['group'] = (fedfunds_grouped_cycles['cycle_filled'] !=␣
 ↪fedfunds_grouped_cycles['cycle_filled'].shift(1)).cumsum()

# Group by both the group key and Cycle label
cycle_ranges = (
    fedfunds_grouped_cycles.groupby(['group', 'cycle_filled'], sort=False)
    .agg(
        start_date=('start_date', 'first'),
        end_date=('end_date', 'last'),
        fed_funds_start=('fed_funds_start', 'first'),
        fed_funds_end=('fed_funds_end', 'last')
    )
    .reset_index(drop=False)
    .drop(columns='group')
)

# Calc change in Fed Funds rate for each cycle
cycle_ranges["fed_funds_change"] = cycle_ranges["fed_funds_end"] -␣
 ↪cycle_ranges["fed_funds_start"]

# Add cycle labels
cycle_labels = [f"Cycle {i+1}" for i in range(len(cycle_ranges))]

# Combine labels with cycle_ranges
cycle_ranges['cycle_label'] = cycle_labels

display(cycle_ranges)
```

```
          cycle_filled start_date   end_date  fed_funds_start  fed_funds_end  \
```

| | | | | | |
|---|---|---|---|---|---|
| 0 | Modified Tightening | 1989-12-31 | 1990-06-30 | 0.08250 | 0.08250 |
| 1 | Easing | 1990-06-30 | 1993-03-31 | 0.08250 | 0.03000 |
| 2 | Modified Tightening | 1993-03-31 | 1994-01-31 | 0.03000 | 0.03000 |
| 3 | Tightening | 1994-01-31 | 1995-06-30 | 0.03000 | 0.06000 |
| 4 | Easing | 1995-06-30 | 1996-07-31 | 0.06000 | 0.05250 |
| 5 | Modified Tightening | 1996-07-31 | 1997-02-28 | 0.05250 | 0.05250 |
| 6 | Tightening | 1997-02-28 | 1997-09-30 | 0.05250 | 0.05500 |
| 7 | Modified Tightening | 1997-09-30 | 1998-08-31 | 0.05500 | 0.05500 |
| 8 | Easing | 1998-08-31 | 1999-05-31 | 0.05500 | 0.04750 |
| 9 | Tightening | 1999-05-31 | 2000-11-30 | 0.04750 | 0.06500 |
| 10 | Modified Tightening | 2000-11-30 | 2000-12-31 | 0.06500 | 0.06500 |
| 11 | Easing | 2000-12-31 | 2002-06-30 | 0.06500 | 0.01750 |
| 12 | Modified Tightening | 2002-06-30 | 2002-10-31 | 0.01750 | 0.01750 |
| 13 | Easing | 2002-10-31 | 2003-12-31 | 0.01750 | 0.01000 |
| 14 | Modified Tightening | 2003-12-31 | 2004-05-31 | 0.01000 | 0.01000 |
| 15 | Tightening | 2004-05-31 | 2006-12-31 | 0.01000 | 0.05250 |
| 16 | Modified Tightening | 2006-12-31 | 2007-08-31 | 0.05250 | 0.05250 |
| 17 | Easing | 2007-08-31 | 2009-07-31 | 0.05250 | 0.00250 |
| 18 | Modified Tightening | 2009-07-31 | 2015-11-30 | 0.00250 | 0.00250 |
| 19 | Tightening | 2015-11-30 | 2016-06-30 | 0.00250 | 0.00500 |
| 20 | Modified Tightening | 2016-06-30 | 2016-11-30 | 0.00500 | 0.00500 |
| 21 | Tightening | 2016-11-30 | 2019-06-30 | 0.00500 | 0.02500 |
| 22 | Modified Tightening | 2019-06-30 | 2019-07-31 | 0.02500 | 0.02500 |
| 23 | Easing | 2019-07-31 | 2020-09-30 | 0.02500 | 0.00250 |
| 24 | Modified Tightening | 2020-09-30 | 2022-02-28 | 0.00250 | 0.00250 |
| 25 | Tightening | 2022-02-28 | 2024-01-31 | 0.00250 | 0.05500 |
| 26 | Modified Tightening | 2024-01-31 | 2024-08-31 | 0.05500 | 0.05500 |
| 27 | Easing | 2024-08-31 | 2025-06-30 | 0.05500 | 0.04500 |
| 28 | Modified Tightening | 2025-06-30 | 2025-08-31 | 0.04500 | 0.04500 |
| 29 | Easing | 2025-08-31 | 2026-01-31 | 0.04500 | 0.03750 |

| | fed_funds_change | cycle_label |
|---|---|---|
| 0 | 0.00000 | Cycle 1 |
| 1 | -0.05250 | Cycle 2 |
| 2 | 0.00000 | Cycle 3 |
| 3 | 0.03000 | Cycle 4 |
| 4 | -0.00750 | Cycle 5 |
| 5 | 0.00000 | Cycle 6 |
| 6 | 0.00250 | Cycle 7 |
| 7 | 0.00000 | Cycle 8 |
| 8 | -0.00750 | Cycle 9 |
| 9 | 0.01750 | Cycle 10 |
| 10 | 0.00000 | Cycle 11 |
| 11 | -0.04750 | Cycle 12 |
| 12 | 0.00000 | Cycle 13 |
| 13 | -0.00750 | Cycle 14 |
| 14 | 0.00000 | Cycle 15 |
| 15 | 0.04250 | Cycle 16 |

```
16           0.00000    Cycle 17
17          -0.05000    Cycle 18
18           0.00000    Cycle 19
19           0.00250    Cycle 20
20           0.00000    Cycle 21
21           0.02000    Cycle 22
22           0.00000    Cycle 23
23          -0.02250    Cycle 24
24           0.00000    Cycle 25
25           0.05250    Cycle 26
26           0.00000    Cycle 27
27          -0.01000    Cycle 28
28           0.00000    Cycle 29
29          -0.00750    Cycle 30
```

## 0.6 Return Performance By Fed Policy Cycle

Moving on, we will now look at the performance of four (4) different asset classes during each Fed cycle. We'll use the following Bloomberg indices:

- SPXT_S&P 500 Total Return Index (Stocks)
- SPBDU10T_S&P US Treasury Bond 7-10 Year Total Return Index (Bonds)
- LF98TRUU_Bloomberg US Corporate High Yield Total Return Index Value Unhedged USD (High Yield Bonds)
- XAU_Gold USD Spot (Gold)

### 0.6.1 Stocks

First, we will clean and load the data for the S&P 500 Total Return Index (SPXT):

[11]:
```
bb_clean_data(
    base_directory=DATA_DIR,
    fund_ticker_name="SPXT_S&P 500 Total Return Index",
    source="Bloomberg",
    asset_class="Indices",
    excel_export=True,
    pickle_export=True,
    output_confirmation=False,
)

spxt = load_data(
    base_directory=DATA_DIR,
    ticker="SPXT_S&P 500 Total Return Index_Clean",
    source="Bloomberg",
    asset_class="Indices",
    timeframe="Daily",
    file_format="pickle",
)
```

```
# Filter SPXT to date range
spxt = spxt[(spxt.index >= pd.to_datetime(start_date)) & (spxt.index <= pd.
 ↪to_datetime(end_date))]

# Drop everything except the "close" column
spxt = spxt[["Close"]]

# Resample to monthly frequency
spxt_monthly = spxt.resample("M").last()
spxt_monthly["Monthly_Return"] = spxt_monthly["Close"].pct_change()
spxt_monthly = spxt_monthly.dropna()

display(spxt_monthly)
```

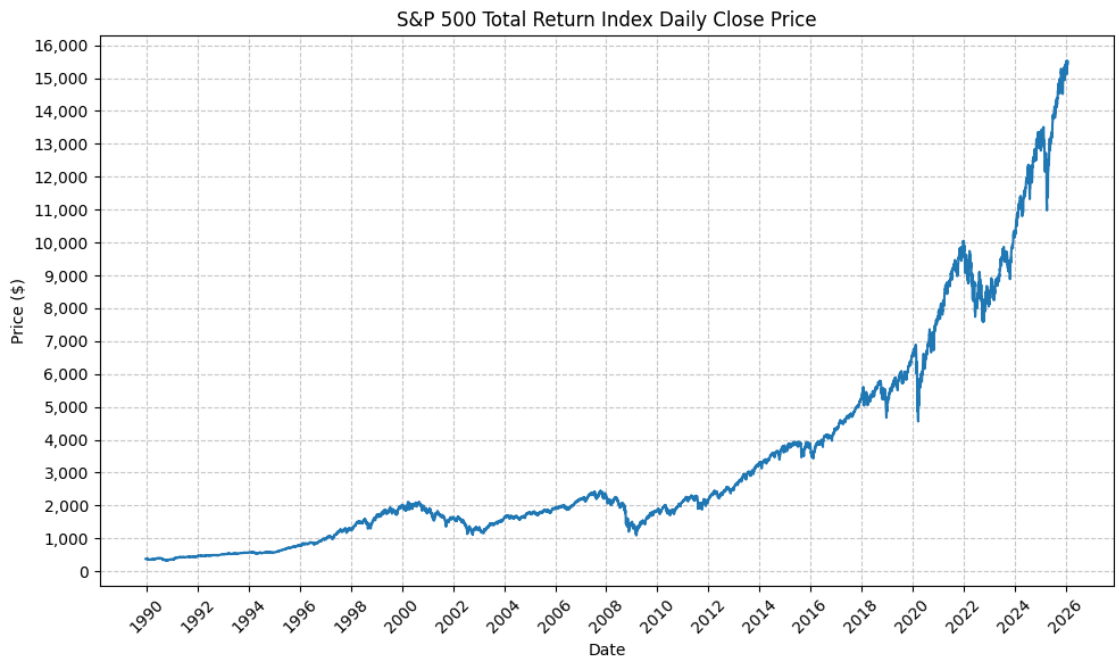
```
                 Close  Monthly_Return
Date
1990-01-31   353.94000        -0.06713
1990-02-28   358.50000         0.01288
1990-03-31         368         0.02650
1990-04-30   358.81000        -0.02497
1990-05-31   393.80000         0.09752
…                 …             …
2025-09-30 14826.80000         0.03650
2025-10-31 15173.95000         0.02341
2025-11-30 15211.14000         0.00245
2025-12-31 15220.45000         0.00061
2026-01-31 15441.15000         0.01450

[433 rows x 2 columns]
```

Next, we can plot the price history before calculating the cycle performance:

[12]:
```
plot_timeseries(
    price_df=spxt,
    plot_start_date=None,
    plot_end_date=None,
    plot_columns=["Close"],
    title="S&P 500 Total Return Index Daily Close Price",
    x_label="Date",
    x_format="Year",
    x_tick_spacing=2,
    x_tick_rotation=45,
    y_label="Price ($)",
    y_format="Decimal",
    y_format_decimal_places=0,
    y_tick_spacing=1000,
    grid=True,
    legend=False,
```

```
    export_plot=False,
    plot_file_name=None,
)
```




Next, we will calculate the performance for SPY based on the pre-defined Fed cycles:

[13]:
```
spxt_cycles = calc_fed_cycle_asset_performance(
    start_date=cycle_ranges["start_date"],
    end_date=cycle_ranges["end_date"],
    label=cycle_ranges["cycle_label"],
    fed_funds_change=cycle_ranges["fed_funds_change"],
    monthly_returns=spxt_monthly,
)

display(spxt_cycles)
```

```
       Cycle       Start         End  Months  CumulativeReturn  \
0    Cycle 1  1989-12-31  1990-06-30       6           0.03092
1    Cycle 2  1990-06-30  1993-03-31      34           0.36800
2    Cycle 3  1993-03-31  1994-01-31      11           0.11359
3    Cycle 4  1994-01-31  1995-06-30      18           0.21800
4    Cycle 5  1995-06-30  1996-07-31      14           0.23230
5    Cycle 6  1996-07-31  1997-02-28       8           0.19592
6    Cycle 7  1997-02-28  1997-09-30       8           0.22017
7    Cycle 8  1997-09-30  1998-08-31      12           0.08094
8    Cycle 9  1998-08-31  1999-05-31      10           0.17553
```

| | | | | | |
|---|---|---|---|---|---|
| 9 | Cycle 10 | 1999-05-31 | 2000-11-30 | 19 | 0.00401 |
| 10 | Cycle 11 | 2000-11-30 | 2000-12-31 | 2 | -0.07433 |
| 11 | Cycle 12 | 2000-12-31 | 2002-06-30 | 19 | -0.23106 |
| 12 | Cycle 13 | 2002-06-30 | 2002-10-31 | 5 | -0.16407 |
| 13 | Cycle 14 | 2002-10-31 | 2003-12-31 | 15 | 0.39543 |
| 14 | Cycle 15 | 2003-12-31 | 2004-05-31 | 6 | 0.06792 |
| 15 | Cycle 16 | 2004-05-31 | 2006-12-31 | 32 | 0.34572 |
| 16 | Cycle 17 | 2006-12-31 | 2007-08-31 | 9 | 0.06671 |
| 17 | Cycle 18 | 2007-08-31 | 2009-07-31 | 24 | -0.28840 |
| 18 | Cycle 19 | 2009-07-31 | 2015-11-30 | 77 | 1.59039 |
| 19 | Cycle 20 | 2015-11-30 | 2016-06-30 | 8 | 0.02502 |
| 20 | Cycle 21 | 2016-06-30 | 2016-11-30 | 6 | 0.06008 |
| 21 | Cycle 22 | 2016-11-30 | 2019-06-30 | 32 | 0.46031 |
| 22 | Cycle 23 | 2019-06-30 | 2019-07-31 | 2 | 0.08586 |
| 23 | Cycle 24 | 2019-07-31 | 2020-09-30 | 15 | 0.17105 |
| 24 | Cycle 25 | 2020-09-30 | 2022-02-28 | 18 | 0.27728 |
| 25 | Cycle 26 | 2022-02-28 | 2024-01-31 | 24 | 0.10892 |
| 26 | Cycle 27 | 2024-01-31 | 2024-08-31 | 8 | 0.19526 |
| 27 | Cycle 28 | 2024-08-31 | 2025-06-30 | 11 | 0.13779 |
| 28 | Cycle 29 | 2025-06-30 | 2025-08-31 | 3 | 0.09622 |
| 29 | Cycle 30 | 2025-08-31 | 2026-01-31 | 6 | 0.10133 |

| | CumulativeReturnPct | AverageMonthlyReturn | AverageMonthlyReturnPct | \ |
|---|---|---|---|---|
| 0 | 3.09164 | 0.00634 | 0.63403 | |
| 1 | 36.80041 | 0.00996 | 0.99573 | |
| 2 | 11.35920 | 0.01001 | 1.00089 | |
| 3 | 21.80042 | 0.01141 | 1.14060 | |
| 4 | 23.23023 | 0.01527 | 1.52704 | |
| 5 | 19.59152 | 0.02336 | 2.33576 | |
| 6 | 22.01713 | 0.02631 | 2.63056 | |
| 7 | 8.09434 | 0.00812 | 0.81242 | |
| 8 | 17.55334 | 0.01849 | 1.84915 | |
| 9 | 0.40140 | 0.00127 | 0.12700 | |
| 10 | -7.43308 | -0.03697 | -3.69725 | |
| 11 | -23.10629 | -0.01254 | -1.25394 | |
| 12 | -16.40672 | -0.03266 | -3.26564 | |
| 13 | 39.54292 | 0.02325 | 2.32521 | |
| 14 | 6.79152 | 0.01127 | 1.12720 | |
| 15 | 34.57166 | 0.00952 | 0.95192 | |
| 16 | 6.67103 | 0.00748 | 0.74817 | |
| 17 | -28.83990 | -0.01191 | -1.19125 | |
| 18 | 159.03905 | 0.01314 | 1.31414 | |
| 19 | 2.50250 | 0.00356 | 0.35617 | |
| 20 | 6.00766 | 0.00997 | 0.99744 | |
| 21 | 46.03091 | 0.01256 | 1.25618 | |
| 22 | 8.58628 | 0.04242 | 4.24249 | |
| 23 | 17.10456 | 0.01234 | 1.23424 | |
| 24 | 27.72844 | 0.01457 | 1.45670 | |

| | | | |
|---|---|---|---|
| 25 | 10.89196 | 0.00584 | 0.58363 |
| 26 | 19.52588 | 0.02293 | 2.29282 |
| 27 | 13.77869 | 0.01244 | 1.24436 |
| 28 | 9.62171 | 0.03119 | 3.11890 |
| 29 | 10.13298 | 0.01629 | 1.62914 |

| | AnnualizedReturn | AnnualizedReturnPct | Volatility | FedFundsChange | \ |
|---|---|---|---|---|---|
| 0 | 0.06279 | 6.27887 | 0.19170 | 0.00000 | |
| 1 | 0.11694 | 11.69426 | 0.13198 | -0.05250 | |
| 2 | 0.12454 | 12.45375 | 0.06918 | 0.00000 | |
| 3 | 0.14051 | 14.05103 | 0.09928 | 0.03000 | |
| 4 | 0.19607 | 19.60729 | 0.07840 | -0.00750 | |
| 5 | 0.30783 | 30.78278 | 0.14361 | 0.00000 | |
| 6 | 0.34782 | 34.78178 | 0.17547 | 0.00250 | |
| 7 | 0.08094 | 8.09434 | 0.20028 | 0.00000 | |
| 8 | 0.21418 | 21.41769 | 0.23543 | -0.00750 | |
| 9 | 0.00253 | 0.25333 | 0.16483 | 0.01750 | |
| 10 | -0.37088 | -37.08781 | 0.20511 | 0.00000 | |
| 11 | -0.15291 | -15.29071 | 0.17313 | -0.04750 | |
| 12 | -0.34955 | -34.95539 | 0.27616 | 0.00000 | |
| 13 | 0.30547 | 30.54681 | 0.14377 | -0.00750 | |
| 14 | 0.14044 | 14.04429 | 0.08737 | 0.00000 | |
| 15 | 0.11778 | 11.77833 | 0.06998 | 0.04250 | |
| 16 | 0.08992 | 8.99217 | 0.08722 | 0.00000 | |
| 17 | -0.15644 | -15.64355 | 0.22888 | -0.05000 | |
| 18 | 0.15990 | 15.99000 | 0.13125 | 0.00000 | |
| 19 | 0.03777 | 3.77714 | 0.11389 | 0.00250 | |
| 20 | 0.12376 | 12.37623 | 0.07708 | 0.00000 | |
| 21 | 0.15257 | 15.25686 | 0.12685 | 0.02000 | |
| 22 | 0.63927 | 63.92672 | 0.13743 | 0.00000 | |
| 23 | 0.13464 | 13.46425 | 0.21140 | -0.02250 | |
| 24 | 0.17722 | 17.72221 | 0.15078 | 0.00000 | |
| 25 | 0.05305 | 5.30525 | 0.19480 | 0.05250 | |
| 26 | 0.30675 | 30.67513 | 0.10238 | 0.00000 | |
| 27 | 0.15122 | 15.12175 | 0.13032 | -0.01000 | |
| 28 | 0.44406 | 44.40637 | 0.05911 | 0.00000 | |
| 29 | 0.21293 | 21.29273 | 0.04688 | -0.00750 | |

| | FedFundsChange_bps | FFR_AnnualizedChange | FFR_AnnualizedChange_bps | \ |
|---|---|---|---|---|
| 0 | 0.00000 | 0.00000 | 0.00000 | |
| 1 | -525.00000 | -0.01853 | -185.29412 | |
| 2 | 0.00000 | 0.00000 | 0.00000 | |
| 3 | 300.00000 | 0.02000 | 200.00000 | |
| 4 | -75.00000 | -0.00643 | -64.28571 | |
| 5 | 0.00000 | 0.00000 | 0.00000 | |
| 6 | 25.00000 | 0.00375 | 37.50000 | |
| 7 | 0.00000 | 0.00000 | 0.00000 | |
| 8 | -75.00000 | -0.00900 | -90.00000 | |

| | | | |
|---|---|---|---|
| 9 | 175.00000 | 0.01105 | 110.52632 |
| 10 | 0.00000 | 0.00000 | 0.00000 |
| 11 | -475.00000 | -0.03000 | -300.00000 |
| 12 | 0.00000 | 0.00000 | 0.00000 |
| 13 | -75.00000 | -0.00600 | -60.00000 |
| 14 | 0.00000 | 0.00000 | 0.00000 |
| 15 | 425.00000 | 0.01594 | 159.37500 |
| 16 | 0.00000 | 0.00000 | 0.00000 |
| 17 | -500.00000 | -0.02500 | -250.00000 |
| 18 | 0.00000 | 0.00000 | 0.00000 |
| 19 | 25.00000 | 0.00375 | 37.50000 |
| 20 | 0.00000 | 0.00000 | 0.00000 |
| 21 | 200.00000 | 0.00750 | 75.00000 |
| 22 | 0.00000 | 0.00000 | 0.00000 |
| 23 | -225.00000 | -0.01800 | -180.00000 |
| 24 | 0.00000 | 0.00000 | 0.00000 |
| 25 | 525.00000 | 0.02625 | 262.50000 |
| 26 | 0.00000 | 0.00000 | 0.00000 |
| 27 | -100.00000 | -0.01091 | -109.09091 |
| 28 | 0.00000 | 0.00000 | 0.00000 |
| 29 | -75.00000 | -0.01500 | -150.00000 |

| | Label |
|---|---|
| 0 | Cycle 1, 1989-12-31 to 1990-06-30 |
| 1 | Cycle 2, 1990-06-30 to 1993-03-31 |
| 2 | Cycle 3, 1993-03-31 to 1994-01-31 |
| 3 | Cycle 4, 1994-01-31 to 1995-06-30 |
| 4 | Cycle 5, 1995-06-30 to 1996-07-31 |
| 5 | Cycle 6, 1996-07-31 to 1997-02-28 |
| 6 | Cycle 7, 1997-02-28 to 1997-09-30 |
| 7 | Cycle 8, 1997-09-30 to 1998-08-31 |
| 8 | Cycle 9, 1998-08-31 to 1999-05-31 |
| 9 | Cycle 10, 1999-05-31 to 2000-11-30 |
| 10 | Cycle 11, 2000-11-30 to 2000-12-31 |
| 11 | Cycle 12, 2000-12-31 to 2002-06-30 |
| 12 | Cycle 13, 2002-06-30 to 2002-10-31 |
| 13 | Cycle 14, 2002-10-31 to 2003-12-31 |
| 14 | Cycle 15, 2003-12-31 to 2004-05-31 |
| 15 | Cycle 16, 2004-05-31 to 2006-12-31 |
| 16 | Cycle 17, 2006-12-31 to 2007-08-31 |
| 17 | Cycle 18, 2007-08-31 to 2009-07-31 |
| 18 | Cycle 19, 2009-07-31 to 2015-11-30 |
| 19 | Cycle 20, 2015-11-30 to 2016-06-30 |
| 20 | Cycle 21, 2016-06-30 to 2016-11-30 |
| 21 | Cycle 22, 2016-11-30 to 2019-06-30 |
| 22 | Cycle 23, 2019-06-30 to 2019-07-31 |
| 23 | Cycle 24, 2019-07-31 to 2020-09-30 |
| 24 | Cycle 25, 2020-09-30 to 2022-02-28 |

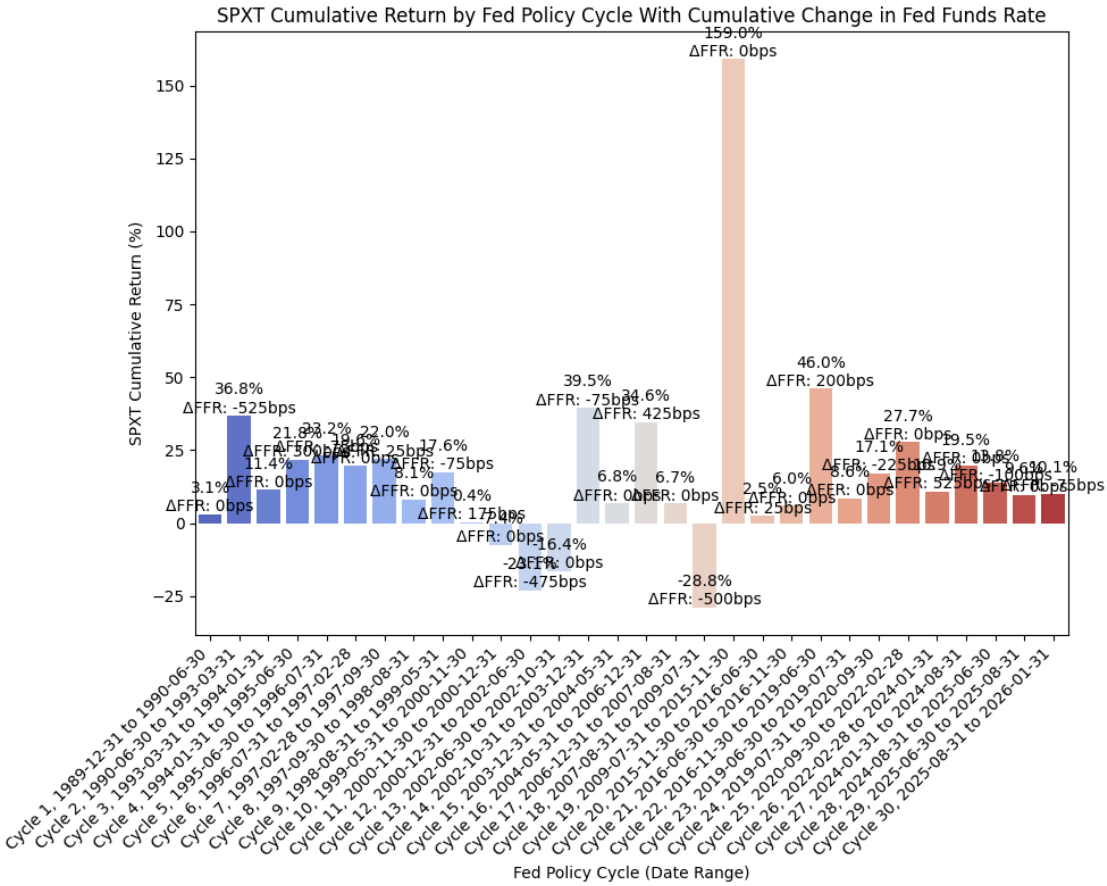
```
25  Cycle 26, 2022-02-28 to 2024-01-31
26  Cycle 27, 2024-01-31 to 2024-08-31
27  Cycle 28, 2024-08-31 to 2025-06-30
28  Cycle 29, 2025-06-30 to 2025-08-31
29  Cycle 30, 2025-08-31 to 2026-01-31
```

This gives us the following data points:

- Cycle start date
- Cycle end date
- Number of months in the cycle
- Cumulative return during the cycle (decimal and percent)
- Average monthly return during the cycle (decimal and percent)
- Annualized return during the cycle (decimal and percent)
- Return volatility during the cycle
- Cumulative change in FFR during the cycle (decimal and basis points)
- Annualized change in FFR during the cycle (decimal and basis points)

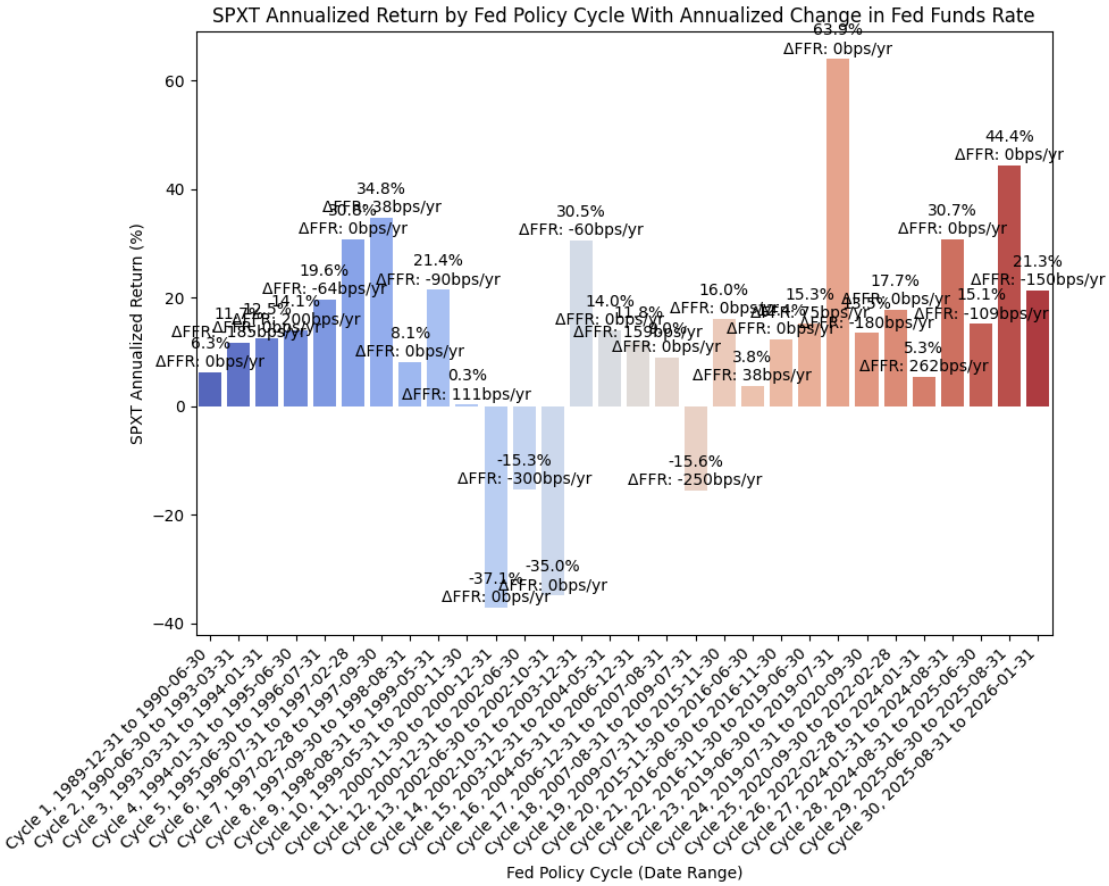
From the above DataFrame, we can then plot the cumulative and annualized returns for each cycle in a bar chart. First, the cumulative returns along with the cumulative change in FFR:

[14]:
```
plot_bar_returns_ffr_change(
    cycle_df=spxt_cycles,
    asset_label="SPXT",
    annualized_or_cumulative="Cumulative",
)
```




And then the annualized returns along with the annualized change in FFR:

[15]:
```
plot_bar_returns_ffr_change(
    cycle_df=spxt_cycles,
    asset_label="SPXT",
    annualized_or_cumulative="Annualized",
)
```




The cumulative returns plot is not particularly insightful, but there are some interesting observations to be gained from the annualized returns plot. During the past two (2) rate cutting cycles (cycles 11/12/13/14 and 18), stocks have exhibited negative returns during the rate cutting cycle. However, after the rate cutting cycle was complete, returns during the following cycle (when rates were usually flat) were quite strong and higher than the historical mean return for the S&P 500. The economic intuition for this behavior is valid; as the economy weakens, investors are concerned about the pricing of equities, the returns become negative, and the Fed responds with cutting rates. The exact timing of when the Fed begins cutting rates is one of the unknowns; the Fed could be ahead of the curve, cutting rates as economic data begins to prompt that action, or behind the curve, where the ecomony rolls over rapidly and even the Fed's actions are not enough to halt the economic contraction.

Finally, we can run an OLS regression to check fit:

[16]:
```
df = spxt_cycles

#=== Don't modify below this line ===

# Run OLS regression with statsmodels
X = df["FFR_AnnualizedChange_bps"]
```

```
y = df["AnnualizedReturnPct"]
X = sm.add_constant(X)
model = sm.OLS(y, X).fit()
print(model.summary())
print(f"Intercept: {model.params[0]}, Slope: {model.params[1]}")  # Intercept␣
↪and slope

# Calc X and Y values for regression line
X_vals = np.linspace(X.min(), X.max(), 100)
Y_vals = model.params[0] + model.params[1] * X_vals
```

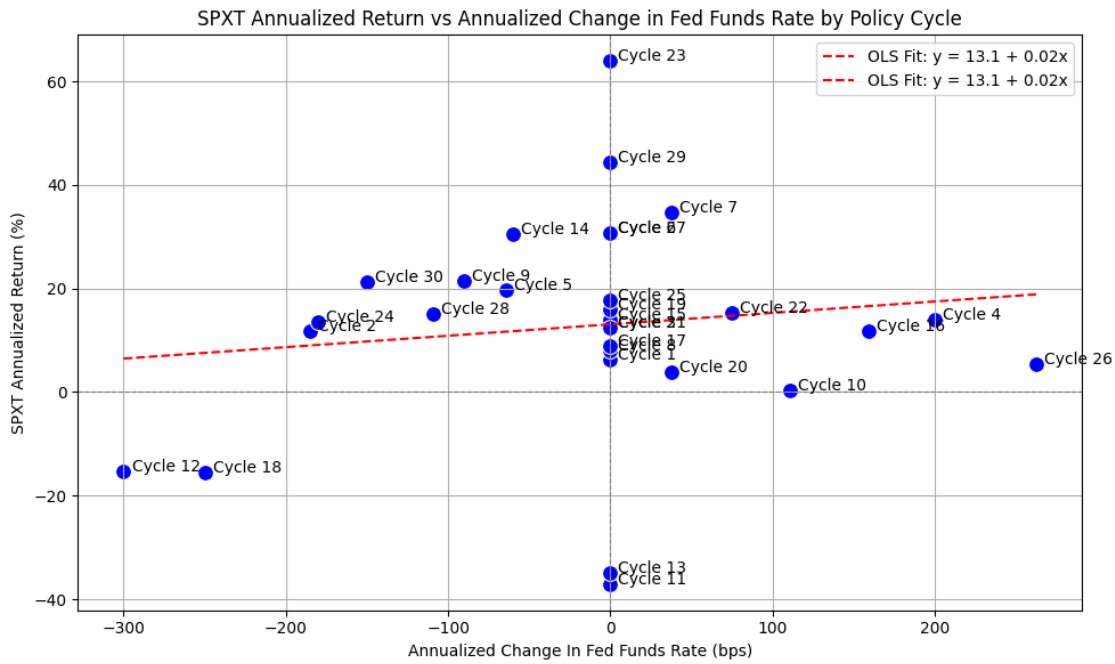
```
                            OLS Regression Results
==============================================================================
Dep. Variable:     AnnualizedReturnPct   R-squared:                       0.017
Model:                             OLS   Adj. R-squared:                 -0.018
Method:                  Least Squares   F-statistic:                    0.4786
Date:                 Tue, 24 Feb 2026   Prob (F-statistic):              0.495
Time:                         13:51:24   Log-Likelihood:                -132.24
No. Observations:                   30   AIC:                             268.5
Df Residuals:                       28   BIC:                             271.3
Df Model:                            1
Covariance Type:             nonrobust
==================================================================================
============
                                coef    std err          t      P>|t|      [0.025
0.975]
----------------------------------------------------------------------------------
------------
const                        13.0760      3.792      3.448      0.002       5.308
20.844
FFR_AnnualizedChange_bps      0.0221      0.032      0.692      0.495      -0.043
0.087
==============================================================================
Omnibus:                        4.070   Durbin-Watson:                   1.523
Prob(Omnibus):                  0.131   Jarque-Bera (JB):                2.894
Skew:                          -0.320   Prob(JB):                        0.235
Kurtosis:                       4.381   Cond. No.                         120.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
Intercept: 13.076015539043272, Slope: 0.022056651501767076
```

And then plot the regression line along with the values:

[17]:
```
plot_scatter_regression_ffr_vs_returns(
    cycle_df=spxt_cycles,
```

```
    asset_label="SPXT",
    x_vals=X_vals,
    y_vals=Y_vals,
    intercept=model.params[0],
    slope=model.params[1],
)
```




Here we can see the data points for cycles 11/12/13/14 and 18 as mentioned above. Interestingly, cycles 28/29/30 (which is the current rate cutting cycle) appears to be an outlier. Of course, the book is not yet finished for cycles 28/29/30, and we could certainly see a bear market in stocks over the next several years.

### 0.6.2 Bonds

Next, we'll run a similar process for medium term bonds using SPBDU10T_S&P US Treasury Bond 7-10 Year Total Return Index.

First, we pull data with the following:

[18]:
```
bb_clean_data(
    base_directory=DATA_DIR,
    fund_ticker_name="SPBDU10T_S&P US Treasury Bond 7-10 Year Total Return␣
 ↪Index",
    source="Bloomberg",
    asset_class="Indices",
    excel_export=True,
```

```
    pickle_export=True,
    output_confirmation=False,
)

treas_10y = load_data(
    base_directory=DATA_DIR,
    ticker="SPBDU10T_S&P US Treasury Bond 7-10 Year Total Return Index_Clean",
    source="Bloomberg",
    asset_class="Indices",
    timeframe="Daily",
    file_format="pickle",
)

# Filter TREAS_10Y to date range
treas_10y = treas_10y[(treas_10y.index >= pd.to_datetime(start_date)) &␣
 ↪(treas_10y.index <= pd.to_datetime(end_date))]

# Drop everything except the "close" column
treas_10y = treas_10y[["Close"]]

# Resample to monthly frequency
treas_10y_monthly = treas_10y.resample("M").last()
treas_10y_monthly["Monthly_Return"] = treas_10y_monthly["Close"].pct_change()
treas_10y_monthly = treas_10y_monthly.dropna()

display(treas_10y_monthly)
```

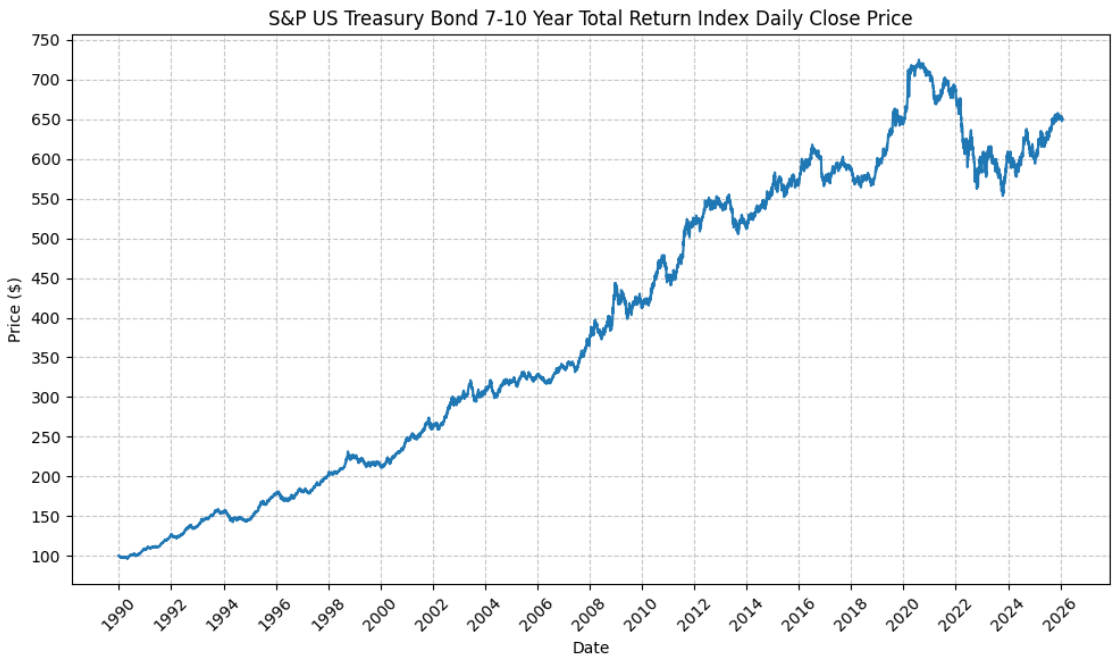
```
               Close  Monthly_Return
Date
1990-01-31  98.01300        -0.01987
1990-02-28  97.99000        -0.00023
1990-03-31  97.98900        -0.00001
1990-04-30  96.60600        -0.01411
1990-05-31  99.64700         0.03148
…               …             …
2025-09-30 645.58400         0.00675
2025-10-31 650.00500         0.00685
2025-11-30 656.63600         0.01020
2025-12-31 652.28800        -0.00662
2026-01-31 650.16900        -0.00325

[433 rows x 2 columns]
```

Next, we can plot the price history before calculating the cycle performance:

[19]:
```
plot_timeseries(
    price_df=treas_10y,
    plot_start_date=None,
```

```
    plot_end_date=None,
    plot_columns=["Close"],
    title="S&P US Treasury Bond 7-10 Year Total Return Index Daily Close Price",
    x_label="Date",
    x_format="Year",
    x_tick_spacing=2,
    x_tick_rotation=45,
    y_label="Price ($)",
    y_format="Decimal",
    y_format_decimal_places=0,
    y_tick_spacing=50,
    grid=True,
    legend=False,
    export_plot=False,
    plot_file_name=None,
)
```




Next, we will calculate the performance for SPY based on the pre-defined Fed cycles:

[20]:
```
treas_10y_cycles = calc_fed_cycle_asset_performance(
    start_date=cycle_ranges["start_date"],
    end_date=cycle_ranges["end_date"],
    label=cycle_ranges["cycle_label"],
    fed_funds_change=cycle_ranges["fed_funds_change"],
    monthly_returns=treas_10y_monthly,
)
```

```
display(treas_10y_cycles)
```

```
       Cycle       Start         End  Months  CumulativeReturn  \
0    Cycle 1  1989-12-31  1990-06-30       6           0.01362
1    Cycle 2  1990-06-30  1993-03-31      34           0.45682
2    Cycle 3  1993-03-31  1994-01-31      11           0.08876
3    Cycle 4  1994-01-31  1995-06-30      18           0.07796
4    Cycle 5  1995-06-30  1996-07-31      14           0.04785
5    Cycle 6  1996-07-31  1997-02-28       8           0.05301
6    Cycle 7  1997-02-28  1997-09-30       8           0.06453
7    Cycle 8  1997-09-30  1998-08-31      12           0.14428
8    Cycle 9  1998-08-31  1999-05-31      10           0.03052
9   Cycle 10  1999-05-31  2000-11-30      19           0.08744
10  Cycle 11  2000-11-30  2000-12-31       2           0.04918
11  Cycle 12  2000-12-31  2002-06-30      19           0.14331
12  Cycle 13  2002-06-30  2002-10-31       5           0.09945
13  Cycle 14  2002-10-31  2003-12-31      15           0.02161
14  Cycle 15  2003-12-31  2004-05-31       6          -0.00088
15  Cycle 16  2004-05-31  2006-12-31      32           0.10267
16  Cycle 17  2006-12-31  2007-08-31       9           0.03300
17  Cycle 18  2007-08-31  2009-07-31      24           0.19843
18  Cycle 19  2009-07-31  2015-11-30      77           0.38995
19  Cycle 20  2015-11-30  2016-06-30       8           0.06790
20  Cycle 21  2016-06-30  2016-11-30       6          -0.03175
21  Cycle 22  2016-11-30  2019-06-30      32           0.05923
22  Cycle 23  2019-06-30  2019-07-31       2           0.01306
23  Cycle 24  2019-07-31  2020-09-30      15           0.13083
24  Cycle 25  2020-09-30  2022-02-28      18          -0.06522
25  Cycle 26  2022-02-28  2024-01-31      24          -0.09878
26  Cycle 27  2024-01-31  2024-08-31       8           0.02748
27  Cycle 28  2024-08-31  2025-06-30      11           0.03294
28  Cycle 29  2025-06-30  2025-08-31       3           0.02670
29  Cycle 30  2025-08-31  2026-01-31       6           0.03081

    CumulativeReturnPct  AverageMonthlyReturn  AverageMonthlyReturnPct  \
0               1.36200               0.00241                  0.24101
1              45.68226               0.01124                  1.12428
2               8.87610               0.00784                  0.78356
3               7.79638               0.00439                  0.43870
4               4.78467               0.00344                  0.34387
5               5.30118               0.00658                  0.65819
6               6.45298               0.00799                  0.79904
7              14.42797               0.01135                  1.13516
8               3.05162               0.00326                  0.32630
9               8.74418               0.00449                  0.44926
10              4.91839               0.02430                  2.42972
11             14.33071               0.00721                  0.72113
```

| | | | |
|---|---|---|---|
| 12 | 9.94496 | 0.01929 | 1.92905 |
| 13 | 2.16130 | 0.00172 | 0.17218 |
| 14 | -0.08831 | 0.00006 | 0.00600 |
| 15 | 10.26670 | 0.00314 | 0.31382 |
| 16 | 3.30047 | 0.00371 | 0.37102 |
| 17 | 19.84327 | 0.00792 | 0.79193 |
| 18 | 38.99485 | 0.00443 | 0.44302 |
| 19 | 6.78998 | 0.00835 | 0.83470 |
| 20 | -3.17490 | -0.00513 | -0.51254 |
| 21 | 5.92338 | 0.00190 | 0.19002 |
| 22 | 1.30568 | 0.00653 | 0.65261 |
| 23 | 13.08268 | 0.00837 | 0.83706 |
| 24 | -6.52177 | -0.00366 | -0.36622 |
| 25 | -9.87840 | -0.00394 | -0.39369 |
| 26 | 2.74817 | 0.00357 | 0.35718 |
| 27 | 3.29408 | 0.00310 | 0.31010 |
| 28 | 2.66978 | 0.00888 | 0.88750 |
| 29 | 3.08068 | 0.00510 | 0.51007 |

| | AnnualizedReturn | AnnualizedReturnPct | Volatility | FedFundsChange | \ |
|---|---|---|---|---|---|
| 0 | 0.02743 | 2.74255 | 0.06657 | 0.00000 | |
| 1 | 0.14202 | 14.20180 | 0.05346 | -0.05250 | |
| 2 | 0.09721 | 9.72108 | 0.04458 | 0.00000 | |
| 3 | 0.05132 | 5.13229 | 0.07287 | 0.03000 | |
| 4 | 0.04087 | 4.08738 | 0.04940 | -0.00750 | |
| 5 | 0.08056 | 8.05623 | 0.05365 | 0.00000 | |
| 6 | 0.09834 | 9.83398 | 0.06285 | 0.00250 | |
| 7 | 0.14428 | 14.42797 | 0.03891 | 0.00000 | |
| 8 | 0.03673 | 3.67303 | 0.08244 | -0.00750 | |
| 9 | 0.05437 | 5.43706 | 0.04250 | 0.01750 | |
| 10 | 0.33386 | 33.38584 | 0.00458 | 0.00000 | |
| 11 | 0.08826 | 8.82645 | 0.05901 | -0.04750 | |
| 12 | 0.25551 | 25.55118 | 0.06682 | 0.00000 | |
| 13 | 0.01725 | 1.72534 | 0.08696 | -0.00750 | |
| 14 | -0.00177 | -0.17654 | 0.07653 | 0.00000 | |
| 15 | 0.03733 | 3.73293 | 0.04440 | 0.04250 | |
| 16 | 0.04425 | 4.42466 | 0.05098 | 0.00000 | |
| 17 | 0.09473 | 9.47295 | 0.09432 | -0.05000 | |
| 18 | 0.05265 | 5.26537 | 0.05945 | 0.00000 | |
| 19 | 0.10356 | 10.35595 | 0.05317 | 0.00250 | |
| 20 | -0.06249 | -6.24899 | 0.08242 | 0.00000 | |
| 21 | 0.02181 | 2.18142 | 0.04975 | 0.02000 | |
| 22 | 0.08094 | 8.09429 | 0.03024 | 0.00000 | |
| 23 | 0.10336 | 10.33591 | 0.06056 | -0.02250 | |
| 24 | -0.04397 | -4.39653 | 0.04430 | 0.00000 | |
| 25 | -0.05068 | -5.06760 | 0.09858 | 0.05250 | |
| 26 | 0.04150 | 4.15045 | 0.06957 | 0.00000 | |
| 27 | 0.03599 | 3.59887 | 0.06280 | -0.01000 | |

| | | | | |
|---|---|---|---|---|
| 28 | 0.11114 | 11.11445 | 0.04409 | 0.00000 |
| 29 | 0.06256 | 6.25626 | 0.02992 | -0.00750 |

| | FedFundsChange_bps | FFR_AnnualizedChange | FFR_AnnualizedChange_bps | \ |
|---|---|---|---|---|
| 0 | 0.00000 | 0.00000 | 0.00000 | |
| 1 | -525.00000 | -0.01853 | -185.29412 | |
| 2 | 0.00000 | 0.00000 | 0.00000 | |
| 3 | 300.00000 | 0.02000 | 200.00000 | |
| 4 | -75.00000 | -0.00643 | -64.28571 | |
| 5 | 0.00000 | 0.00000 | 0.00000 | |
| 6 | 25.00000 | 0.00375 | 37.50000 | |
| 7 | 0.00000 | 0.00000 | 0.00000 | |
| 8 | -75.00000 | -0.00900 | -90.00000 | |
| 9 | 175.00000 | 0.01105 | 110.52632 | |
| 10 | 0.00000 | 0.00000 | 0.00000 | |
| 11 | -475.00000 | -0.03000 | -300.00000 | |
| 12 | 0.00000 | 0.00000 | 0.00000 | |
| 13 | -75.00000 | -0.00600 | -60.00000 | |
| 14 | 0.00000 | 0.00000 | 0.00000 | |
| 15 | 425.00000 | 0.01594 | 159.37500 | |
| 16 | 0.00000 | 0.00000 | 0.00000 | |
| 17 | -500.00000 | -0.02500 | -250.00000 | |
| 18 | 0.00000 | 0.00000 | 0.00000 | |
| 19 | 25.00000 | 0.00375 | 37.50000 | |
| 20 | 0.00000 | 0.00000 | 0.00000 | |
| 21 | 200.00000 | 0.00750 | 75.00000 | |
| 22 | 0.00000 | 0.00000 | 0.00000 | |
| 23 | -225.00000 | -0.01800 | -180.00000 | |
| 24 | 0.00000 | 0.00000 | 0.00000 | |
| 25 | 525.00000 | 0.02625 | 262.50000 | |
| 26 | 0.00000 | 0.00000 | 0.00000 | |
| 27 | -100.00000 | -0.01091 | -109.09091 | |
| 28 | 0.00000 | 0.00000 | 0.00000 | |
| 29 | -75.00000 | -0.01500 | -150.00000 | |

| | Label |
|---|---|
| 0 | Cycle 1, 1989-12-31 to 1990-06-30 |
| 1 | Cycle 2, 1990-06-30 to 1993-03-31 |
| 2 | Cycle 3, 1993-03-31 to 1994-01-31 |
| 3 | Cycle 4, 1994-01-31 to 1995-06-30 |
| 4 | Cycle 5, 1995-06-30 to 1996-07-31 |
| 5 | Cycle 6, 1996-07-31 to 1997-02-28 |
| 6 | Cycle 7, 1997-02-28 to 1997-09-30 |
| 7 | Cycle 8, 1997-09-30 to 1998-08-31 |
| 8 | Cycle 9, 1998-08-31 to 1999-05-31 |
| 9 | Cycle 10, 1999-05-31 to 2000-11-30 |
| 10 | Cycle 11, 2000-11-30 to 2000-12-31 |
| 11 | Cycle 12, 2000-12-31 to 2002-06-30 |

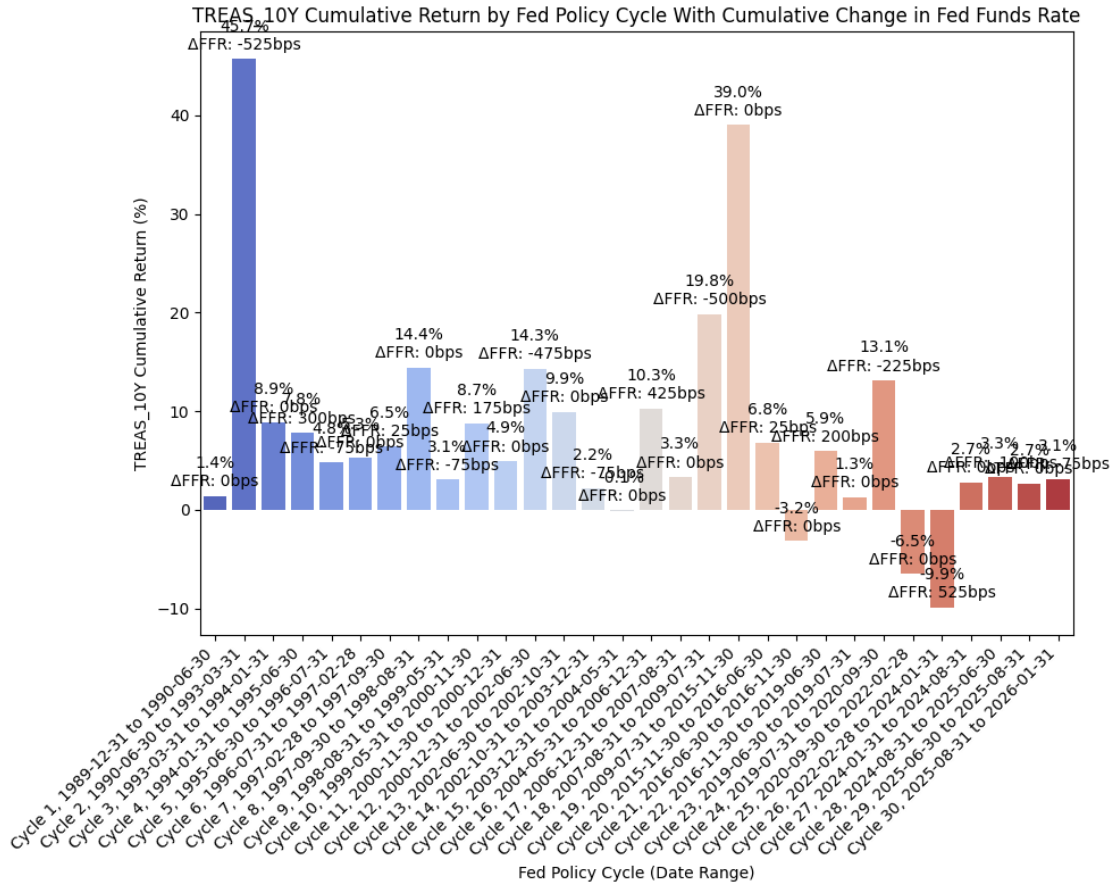
```
12  Cycle 13, 2002-06-30 to 2002-10-31
13  Cycle 14, 2002-10-31 to 2003-12-31
14  Cycle 15, 2003-12-31 to 2004-05-31
15  Cycle 16, 2004-05-31 to 2006-12-31
16  Cycle 17, 2006-12-31 to 2007-08-31
17  Cycle 18, 2007-08-31 to 2009-07-31
18  Cycle 19, 2009-07-31 to 2015-11-30
19  Cycle 20, 2015-11-30 to 2016-06-30
20  Cycle 21, 2016-06-30 to 2016-11-30
21  Cycle 22, 2016-11-30 to 2019-06-30
22  Cycle 23, 2019-06-30 to 2019-07-31
23  Cycle 24, 2019-07-31 to 2020-09-30
24  Cycle 25, 2020-09-30 to 2022-02-28
25  Cycle 26, 2022-02-28 to 2024-01-31
26  Cycle 27, 2024-01-31 to 2024-08-31
27  Cycle 28, 2024-08-31 to 2025-06-30
28  Cycle 29, 2025-06-30 to 2025-08-31
29  Cycle 30, 2025-08-31 to 2026-01-31
```

This gives us the following data points:

- Cycle start date
- Cycle end date
- Number of months in the cycle
- Cumulative return during the cycle (decimal and percent)
- Average monthly return during the cycle (decimal and percent)
- Annualized return during the cycle (decimal and percent)
- Return volatility during the cycle
- Cumulative change in FFR during the cycle (decimal and basis points)
- Annualized change in FFR during the cycle (decimal and basis points)

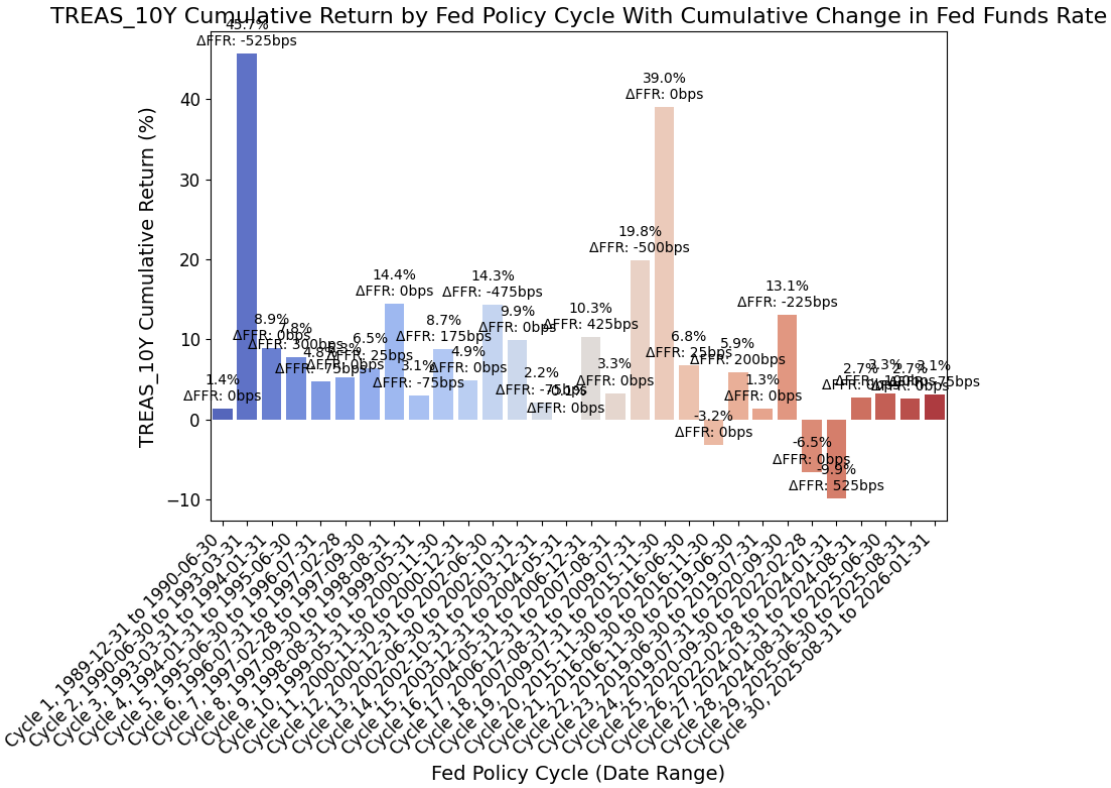
From the above DataFrame, we can then plot the cumulative and annualized returns for each cycle in a bar chart. First, the cumulative returns along with the cumulative change in FFR:

[21]:
```
plot_bar_returns_ffr_change(
    cycle_df=treas_10y_cycles,
    asset_label="TREAS_10Y",
    annualized_or_cumulative="Cumulative",
)
```




And then the annualized returns along with the annualized change in FFR:

[22]:
```
plot_bar_returns_ffr_change(
    cycle_df=treas_10y_cycles,
    asset_label="TREAS_10Y",
    annualized_or_cumulative="Annualized",
)
```




For this dataset, we have cycles 11/12/13/14 exhibiting strong returns, which is consistent with the economic intuition that bonds should perform well during periods of economic weakness and rate cuts. We also see this outcome with cycle 18. It becomes a little more interesting during cycles 25 and 26, where the correlations of stocks and bond returns seemed to align, so we see negative bond returns there. Finally, cycles 28/29/30 also exhibit positive bond returns, which is consistent with our thesis that bonds should perform well during periods of economic weakness and rate cuts.

Finally, we can run an OLS regression to check fit:

[23]:
```
df = treas_10y_cycles

#=== Don't modify below this line ===

# Run OLS regression with statsmodels
X = df["FFR_AnnualizedChange_bps"]
y = df["AnnualizedReturnPct"]
X = sm.add_constant(X)
model = sm.OLS(y, X).fit()
print(model.summary())
print(f"Intercept: {model.params[0]}, Slope: {model.params[1]}")  # Intercept␣
↪and slope

# Calc X and Y values for regression line
```

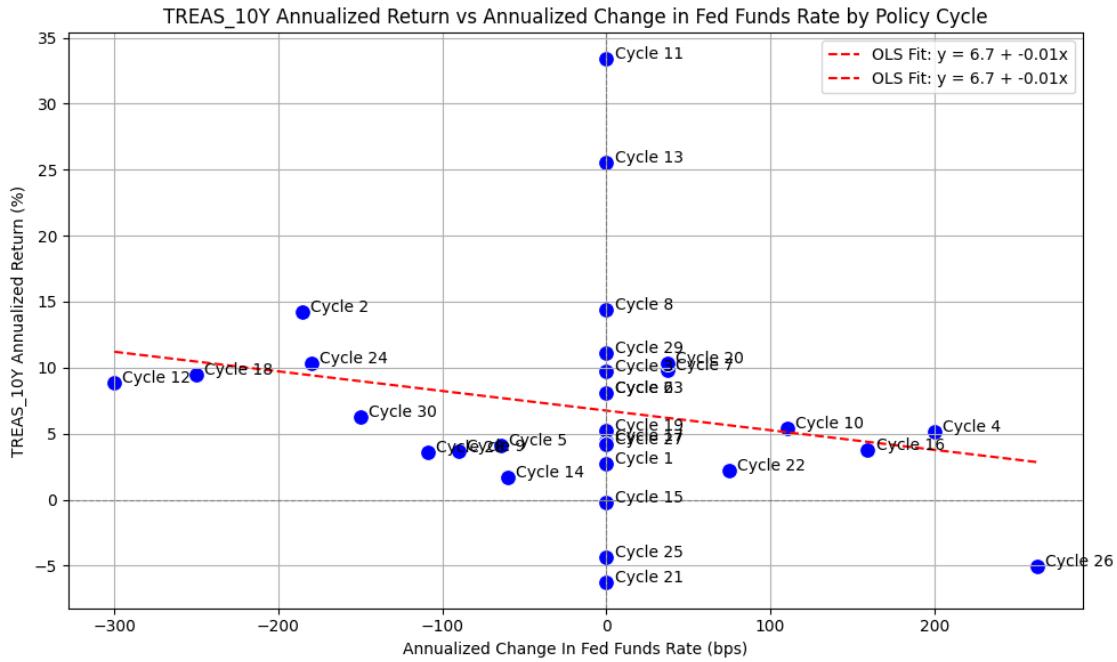
```
X_vals = np.linspace(X.min(), X.max(), 100)
Y_vals = model.params[0] + model.params[1] * X_vals
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:     AnnualizedReturnPct   R-squared:                       0.050
Model:                             OLS   Adj. R-squared:                  0.016
Method:                  Least Squares   F-statistic:                     1.462
Date:                 Tue, 24 Feb 2026   Prob (F-statistic):              0.237
Time:                         13:51:27   Log-Likelihood:                -103.64
No. Observations:                   30   AIC:                             211.3
Df Residuals:                       28   BIC:                             214.1
Df Model:                            1
Covariance Type:             nonrobust
==================================================================================
============
                              coef    std err          t      P>|t|      [0.025
0.975]
----------------------------------------------------------------------------------
------------
const                       6.7457      1.462      4.613      0.000       3.751
9.741
FFR_AnnualizedChange_bps   -0.0149      0.012     -1.209      0.237      -0.040
0.010
==============================================================================
Omnibus:                       19.978   Durbin-Watson:                   2.116
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               29.185
Skew:                           1.582   Prob(JB):                     4.60e-07
Kurtosis:                       6.651   Cond. No.                         120.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
Intercept: 6.745718627069434, Slope: -0.01486255000436152
```

And then plot the regression line along with the values:

[24]:
```
plot_scatter_regression_ffr_vs_returns(
    cycle_df=treas_10y_cycles,
    asset_label="TREAS_10Y",
    x_vals=X_vals,
    y_vals=Y_vals,
    intercept=model.params[0],
    slope=model.params[1],
)
```




The above plot is intriguing because of how well the OLS regression appears to fit the data. It certainly appears that during rate-cutting cycles, bonds are an asset that performs well.

### 0.6.3 High Yield Bonds

Next, we'll run a similar process for high yield bonds using LF98TRUU_Bloomberg US Corporate High Yield Total Return Index Value Unhedged USD.

First, we pull data with the following:

[25]:
```
bb_clean_data(
    base_directory=DATA_DIR,
    fund_ticker_name="LF98TRUU_Bloomberg US Corporate High Yield Total Return␣
 ↪Index Value Unhedged USD",
    source="Bloomberg",
    asset_class="Indices",
    excel_export=True,
    pickle_export=True,
    output_confirmation=False,
)

hy_bonds = load_data(
    base_directory=DATA_DIR,
    ticker="LF98TRUU_Bloomberg US Corporate High Yield Total Return Index Value␣
 ↪Unhedged USD_Clean",
    source="Bloomberg",
```

```
    asset_class="Indices",
    timeframe="Daily",
    file_format="pickle",
)

# Filter HY_BONDS to date range
hy_bonds = hy_bonds[(hy_bonds.index >= pd.to_datetime(start_date)) & (hy_bonds.
 ↪index <= pd.to_datetime(end_date))]

# Drop everything except the "close" column
hy_bonds = hy_bonds[["Close"]]

# Resample to monthly frequency
hy_bonds_monthly = hy_bonds.resample("M").last()
hy_bonds_monthly["Monthly_Return"] = hy_bonds_monthly["Close"].pct_change()
hy_bonds_monthly = hy_bonds_monthly.dropna()

display(hy_bonds_monthly)
```

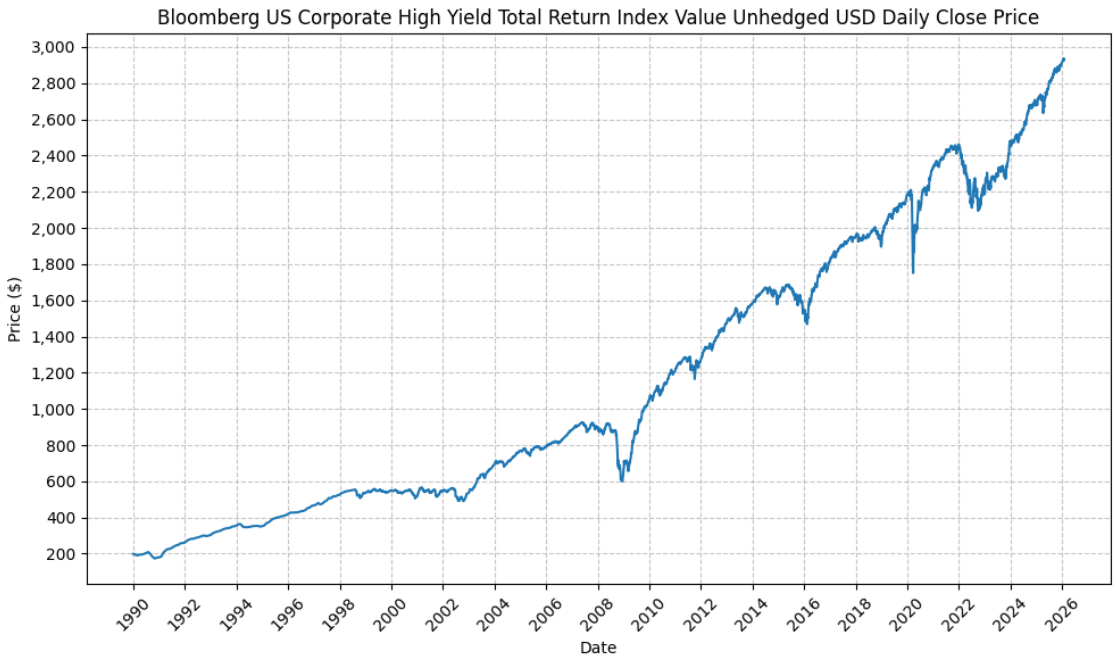
```
                Close  Monthly_Return
Date
1990-01-31  194.21000        -0.02146
1990-02-28  190.20000        -0.02065
1990-03-31  195.19000         0.02624
1990-04-30  194.86000        -0.00169
1990-05-31  198.62000         0.01930
…                …             …
2025-09-30 2876.85000         0.00816
2025-10-31 2881.38000         0.00157
2025-11-30 2898.07000         0.00579
2025-12-31 2914.49000         0.00567
2026-01-31 2929.32000         0.00509

[433 rows x 2 columns]
```

Next, we can plot the price history before calculating the cycle performance:

[26]:
```
plot_timeseries(
    price_df=hy_bonds,
    plot_start_date=None,
    plot_end_date=None,
    plot_columns=["Close"],
    title="Bloomberg US Corporate High Yield Total Return Index Value Unhedged␣
 ↪USD Daily Close Price",
    x_label="Date",
    x_format="Year",
    x_tick_spacing=2,
    x_tick_rotation=45,
```

```
    y_label="Price ($)",
    y_format="Decimal",
    y_format_decimal_places=0,
    y_tick_spacing=200,
    grid=True,
    legend=False,
    export_plot=False,
    plot_file_name=None,
)
```




Next, we will calculate the performance of high yield bonds based on the pre-defined Fed cycles:

[27]:
```
hy_bonds_cycles = calc_fed_cycle_asset_performance(
    start_date=cycle_ranges["start_date"],
    end_date=cycle_ranges["end_date"],
    label=cycle_ranges["cycle_label"],
    fed_funds_change=cycle_ranges["fed_funds_change"],
    monthly_returns=hy_bonds_monthly,
)

display(hy_bonds_cycles)
```

```
       Cycle       Start         End  Months  CumulativeReturn  \
0    Cycle 1  1989-12-31  1990-06-30       6           0.02494
1    Cycle 2  1990-06-30  1993-03-31      34           0.62149
2    Cycle 3  1993-03-31  1994-01-31      11           0.14262
```

| | | | | | |
|---|---|---|---|---|---|
| 3 | Cycle 4 | 1994-01-31 | 1995-06-30 | 18 | 0.11254 |
| 4 | Cycle 5 | 1995-06-30 | 1996-07-31 | 14 | 0.10892 |
| 5 | Cycle 6 | 1996-07-31 | 1997-02-28 | 8 | 0.10479 |
| 6 | Cycle 7 | 1997-02-28 | 1997-09-30 | 8 | 0.09557 |
| 7 | Cycle 8 | 1997-09-30 | 1998-08-31 | 12 | 0.03219 |
| 8 | Cycle 9 | 1998-08-31 | 1999-05-31 | 10 | -0.00728 |
| 9 | Cycle 10 | 1999-05-31 | 2000-11-30 | 19 | -0.08916 |
| 10 | Cycle 11 | 2000-11-30 | 2000-12-31 | 2 | -0.02105 |
| 11 | Cycle 12 | 2000-12-31 | 2002-06-30 | 19 | 0.02117 |
| 12 | Cycle 13 | 2002-06-30 | 2002-10-31 | 5 | -0.10874 |
| 13 | Cycle 14 | 2002-10-31 | 2003-12-31 | 15 | 0.37662 |
| 14 | Cycle 15 | 2003-12-31 | 2004-05-31 | 6 | 0.02188 |
| 15 | Cycle 16 | 2004-05-31 | 2006-12-31 | 32 | 0.25625 |
| 16 | Cycle 17 | 2006-12-31 | 2007-08-31 | 9 | 0.01683 |
| 17 | Cycle 18 | 2007-08-31 | 2009-07-31 | 24 | 0.04908 |
| 18 | Cycle 19 | 2009-07-31 | 2015-11-30 | 77 | 0.83146 |
| 19 | Cycle 20 | 2015-11-30 | 2016-06-30 | 8 | 0.03948 |
| 20 | Cycle 21 | 2016-06-30 | 2016-11-30 | 6 | 0.06426 |
| 21 | Cycle 22 | 2016-11-30 | 2019-06-30 | 32 | 0.17306 |
| 22 | Cycle 23 | 2019-06-30 | 2019-07-31 | 2 | 0.02856 |
| 23 | Cycle 24 | 2019-07-31 | 2020-09-30 | 15 | 0.04631 |
| 24 | Cycle 25 | 2020-09-30 | 2022-02-28 | 18 | 0.06775 |
| 25 | Cycle 26 | 2022-02-28 | 2024-01-31 | 24 | 0.03580 |
| 26 | Cycle 27 | 2024-01-31 | 2024-08-31 | 8 | 0.06285 |
| 27 | Cycle 28 | 2024-08-31 | 2025-06-30 | 11 | 0.08182 |
| 28 | Cycle 29 | 2025-06-30 | 2025-08-31 | 3 | 0.03577 |
| 29 | Cycle 30 | 2025-08-31 | 2026-01-31 | 6 | 0.03937 |

| | CumulativeReturnPct | AverageMonthlyReturn | AverageMonthlyReturnPct | \ |
|---|---|---|---|---|
| 0 | 2.49408 | 0.00432 | 0.43159 | |
| 1 | 62.14883 | 0.01480 | 1.47959 | |
| 2 | 14.26235 | 0.01221 | 1.22129 | |
| 3 | 11.25422 | 0.00607 | 0.60701 | |
| 4 | 10.89235 | 0.00743 | 0.74262 | |
| 5 | 10.47889 | 0.01255 | 1.25548 | |
| 6 | 9.55703 | 0.01156 | 1.15581 | |
| 7 | 3.21938 | 0.00282 | 0.28169 | |
| 8 | -0.72813 | -0.00042 | -0.04217 | |
| 9 | -8.91594 | -0.00480 | -0.47963 | |
| 10 | -2.10510 | -0.01014 | -1.01430 | |
| 11 | 2.11701 | 0.00167 | 0.16693 | |
| 12 | -10.87431 | -0.02215 | -2.21511 | |
| 13 | 37.66182 | 0.02172 | 2.17230 | |
| 14 | 2.18782 | 0.00371 | 0.37115 | |
| 15 | 25.62459 | 0.00722 | 0.72211 | |
| 16 | 1.68350 | 0.00199 | 0.19914 | |
| 17 | 4.90805 | 0.00371 | 0.37092 | |
| 18 | 83.14599 | 0.00808 | 0.80820 | |

| | | | |
|---|---|---|---|
| 19 | 3.94781 | 0.00515 | 0.51508 |
| 20 | 6.42628 | 0.01049 | 1.04900 |
| 21 | 17.30568 | 0.00508 | 0.50759 |
| 22 | 2.85572 | 0.01421 | 1.42144 |
| 23 | 4.63140 | 0.00372 | 0.37244 |
| 24 | 6.77496 | 0.00374 | 0.37444 |
| 25 | 3.58042 | 0.00186 | 0.18605 |
| 26 | 6.28480 | 0.00769 | 0.76866 |
| 27 | 8.18158 | 0.00722 | 0.72241 |
| 28 | 3.57673 | 0.01180 | 1.17991 |
| 29 | 3.93741 | 0.00646 | 0.64627 |

| | AnnualizedReturn | AnnualizedReturnPct | Volatility | FedFundsChange | \ |
|---|---|---|---|---|---|
| 0 | 0.05050 | 5.05036 | 0.07625 | 0.00000 | |
| 1 | 0.18601 | 18.60069 | 0.10898 | -0.05250 | |
| 2 | 0.15656 | 15.65571 | 0.02227 | 0.00000 | |
| 3 | 0.07369 | 7.36869 | 0.05691 | 0.03000 | |
| 4 | 0.09267 | 9.26651 | 0.01906 | -0.00750 | |
| 5 | 0.16123 | 16.12319 | 0.02366 | 0.00000 | |
| 6 | 0.14673 | 14.67279 | 0.04792 | 0.00250 | |
| 7 | 0.03219 | 3.21938 | 0.06631 | 0.00000 | |
| 8 | -0.00873 | -0.87312 | 0.09026 | -0.00750 | |
| 9 | -0.05728 | -5.72758 | 0.05172 | 0.01750 | |
| 10 | -0.11984 | -11.98427 | 0.14432 | 0.00000 | |
| 11 | 0.01332 | 1.33189 | 0.11914 | -0.04750 | |
| 12 | -0.24141 | -24.14102 | 0.13364 | 0.00000 | |
| 13 | 0.29137 | 29.13703 | 0.07004 | -0.00750 | |
| 14 | 0.04424 | 4.42350 | 0.05322 | 0.00000 | |
| 15 | 0.08931 | 8.93138 | 0.04056 | 0.04250 | |
| 16 | 0.02251 | 2.25094 | 0.05992 | 0.00000 | |
| 17 | 0.02425 | 2.42463 | 0.20404 | -0.05000 | |
| 18 | 0.09889 | 9.88931 | 0.06867 | 0.00000 | |
| 19 | 0.05980 | 5.97977 | 0.09107 | 0.00250 | |
| 20 | 0.13266 | 13.26554 | 0.04025 | 0.00000 | |
| 21 | 0.06168 | 6.16825 | 0.04347 | 0.02000 | |
| 22 | 0.18405 | 18.40520 | 0.04203 | 0.00000 | |
| 23 | 0.03688 | 3.68827 | 0.13113 | -0.02250 | |
| 24 | 0.04467 | 4.46712 | 0.04952 | 0.00000 | |
| 25 | 0.01774 | 1.77447 | 0.09916 | 0.05250 | |
| 26 | 0.09574 | 9.57381 | 0.03255 | 0.00000 | |
| 27 | 0.08958 | 8.95776 | 0.03617 | -0.01000 | |
| 28 | 0.15093 | 15.09297 | 0.02414 | 0.00000 | |
| 29 | 0.08030 | 8.02985 | 0.01260 | -0.00750 | |

| | FedFundsChange_bps | FFR_AnnualizedChange | FFR_AnnualizedChange_bps | \ |
|---|---|---|---|---|
| 0 | 0.00000 | 0.00000 | 0.00000 | |
| 1 | -525.00000 | -0.01853 | -185.29412 | |
| 2 | 0.00000 | 0.00000 | 0.00000 | |

| | | | |
|---|---|---|---|
| 3 | 300.00000 | 0.02000 | 200.00000 |
| 4 | -75.00000 | -0.00643 | -64.28571 |
| 5 | 0.00000 | 0.00000 | 0.00000 |
| 6 | 25.00000 | 0.00375 | 37.50000 |
| 7 | 0.00000 | 0.00000 | 0.00000 |
| 8 | -75.00000 | -0.00900 | -90.00000 |
| 9 | 175.00000 | 0.01105 | 110.52632 |
| 10 | 0.00000 | 0.00000 | 0.00000 |
| 11 | -475.00000 | -0.03000 | -300.00000 |
| 12 | 0.00000 | 0.00000 | 0.00000 |
| 13 | -75.00000 | -0.00600 | -60.00000 |
| 14 | 0.00000 | 0.00000 | 0.00000 |
| 15 | 425.00000 | 0.01594 | 159.37500 |
| 16 | 0.00000 | 0.00000 | 0.00000 |
| 17 | -500.00000 | -0.02500 | -250.00000 |
| 18 | 0.00000 | 0.00000 | 0.00000 |
| 19 | 25.00000 | 0.00375 | 37.50000 |
| 20 | 0.00000 | 0.00000 | 0.00000 |
| 21 | 200.00000 | 0.00750 | 75.00000 |
| 22 | 0.00000 | 0.00000 | 0.00000 |
| 23 | -225.00000 | -0.01800 | -180.00000 |
| 24 | 0.00000 | 0.00000 | 0.00000 |
| 25 | 525.00000 | 0.02625 | 262.50000 |
| 26 | 0.00000 | 0.00000 | 0.00000 |
| 27 | -100.00000 | -0.01091 | -109.09091 |
| 28 | 0.00000 | 0.00000 | 0.00000 |
| 29 | -75.00000 | -0.01500 | -150.00000 |

| | Label |
|---|---|
| 0 | Cycle 1, 1989-12-31 to 1990-06-30 |
| 1 | Cycle 2, 1990-06-30 to 1993-03-31 |
| 2 | Cycle 3, 1993-03-31 to 1994-01-31 |
| 3 | Cycle 4, 1994-01-31 to 1995-06-30 |
| 4 | Cycle 5, 1995-06-30 to 1996-07-31 |
| 5 | Cycle 6, 1996-07-31 to 1997-02-28 |
| 6 | Cycle 7, 1997-02-28 to 1997-09-30 |
| 7 | Cycle 8, 1997-09-30 to 1998-08-31 |
| 8 | Cycle 9, 1998-08-31 to 1999-05-31 |
| 9 | Cycle 10, 1999-05-31 to 2000-11-30 |
| 10 | Cycle 11, 2000-11-30 to 2000-12-31 |
| 11 | Cycle 12, 2000-12-31 to 2002-06-30 |
| 12 | Cycle 13, 2002-06-30 to 2002-10-31 |
| 13 | Cycle 14, 2002-10-31 to 2003-12-31 |
| 14 | Cycle 15, 2003-12-31 to 2004-05-31 |
| 15 | Cycle 16, 2004-05-31 to 2006-12-31 |
| 16 | Cycle 17, 2006-12-31 to 2007-08-31 |
| 17 | Cycle 18, 2007-08-31 to 2009-07-31 |
| 18 | Cycle 19, 2009-07-31 to 2015-11-30 |

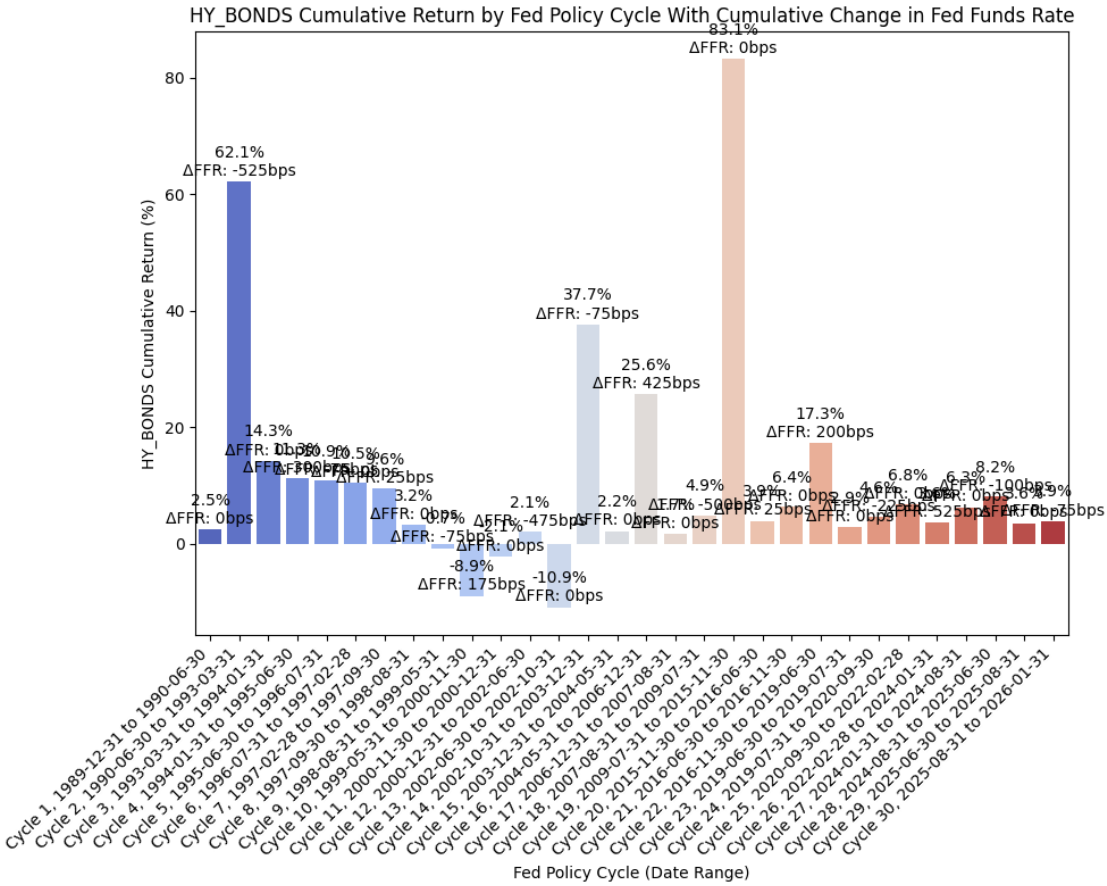
```
19  Cycle 20, 2015-11-30 to 2016-06-30
20  Cycle 21, 2016-06-30 to 2016-11-30
21  Cycle 22, 2016-11-30 to 2019-06-30
22  Cycle 23, 2019-06-30 to 2019-07-31
23  Cycle 24, 2019-07-31 to 2020-09-30
24  Cycle 25, 2020-09-30 to 2022-02-28
25  Cycle 26, 2022-02-28 to 2024-01-31
26  Cycle 27, 2024-01-31 to 2024-08-31
27  Cycle 28, 2024-08-31 to 2025-06-30
28  Cycle 29, 2025-06-30 to 2025-08-31
29  Cycle 30, 2025-08-31 to 2026-01-31
```

This gives us the following data points:

- Cycle start date
- Cycle end date
- Number of months in the cycle
- Cumulative return during the cycle (decimal and percent)
- Average monthly return during the cycle (decimal and percent)
- Annualized return during the cycle (decimal and percent)
- Return volatility during the cycle
- Cumulative change in FFR during the cycle (decimal and basis points)
- Annualized change in FFR during the cycle (decimal and basis points)

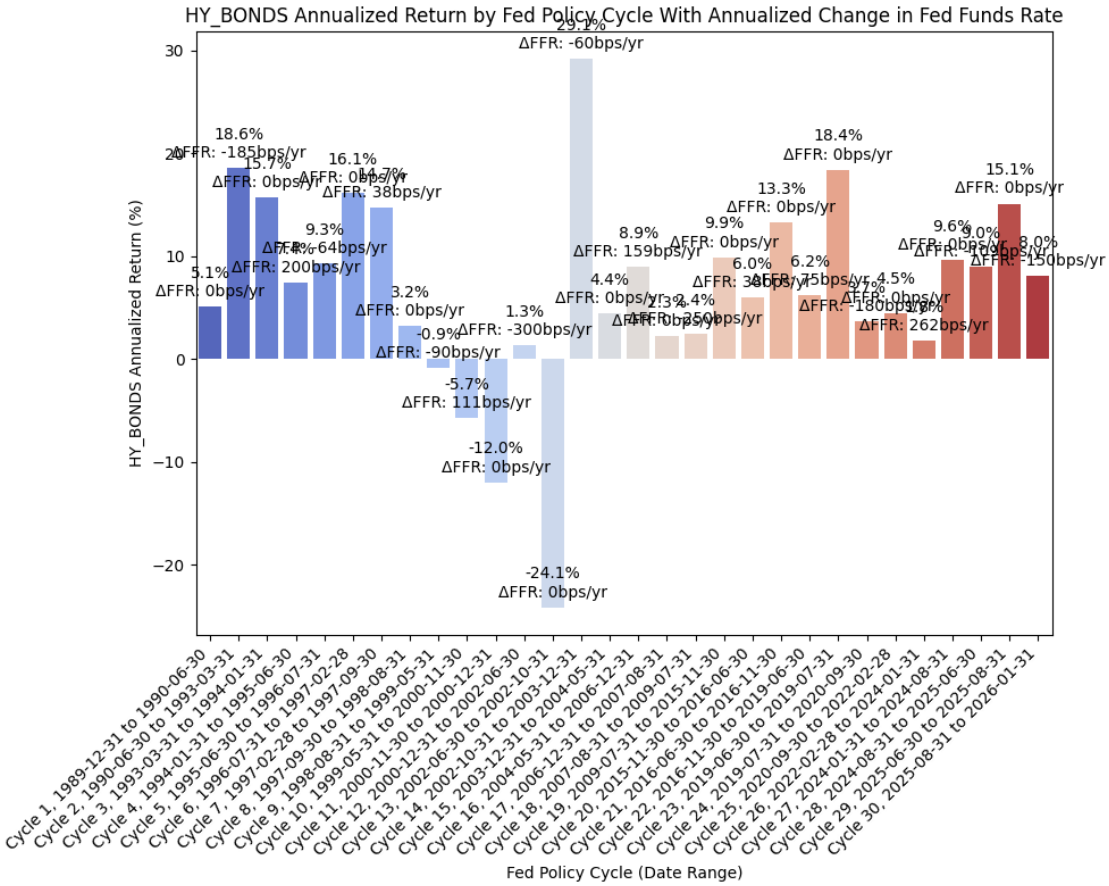
From the above DataFrame, we can then plot the cumulative and annualized returns for each cycle in a bar chart. First, the cumulative returns along with the cumulative change in FFR:

[28]:
```
plot_bar_returns_ffr_change(
    cycle_df=hy_bonds_cycles,
    asset_label="HY_BONDS",
    annualized_or_cumulative="Cumulative",
)
```




And then the annualized returns along with the annualized change in FFR:

[29]:
```
plot_bar_returns_ffr_change(
    cycle_df=hy_bonds_cycles,
    asset_label="HY_BONDS",
    annualized_or_cumulative="Annualized",
)
```




Finally, we can run an OLS regression to check fit:

[30]:
```
df = hy_bonds_cycles

#=== Don't modify below this line ===

# Run OLS regression with statsmodels
X = df["FFR_AnnualizedChange_bps"]
y = df["AnnualizedReturnPct"]
X = sm.add_constant(X)
model = sm.OLS(y, X).fit()
print(model.summary())
print(f"Intercept: {model.params[0]}, Slope: {model.params[1]}")  # Intercept␣
↪and slope

# Calc X and Y values for regression line
X_vals = np.linspace(X.min(), X.max(), 100)
Y_vals = model.params[0] + model.params[1] * X_vals
```

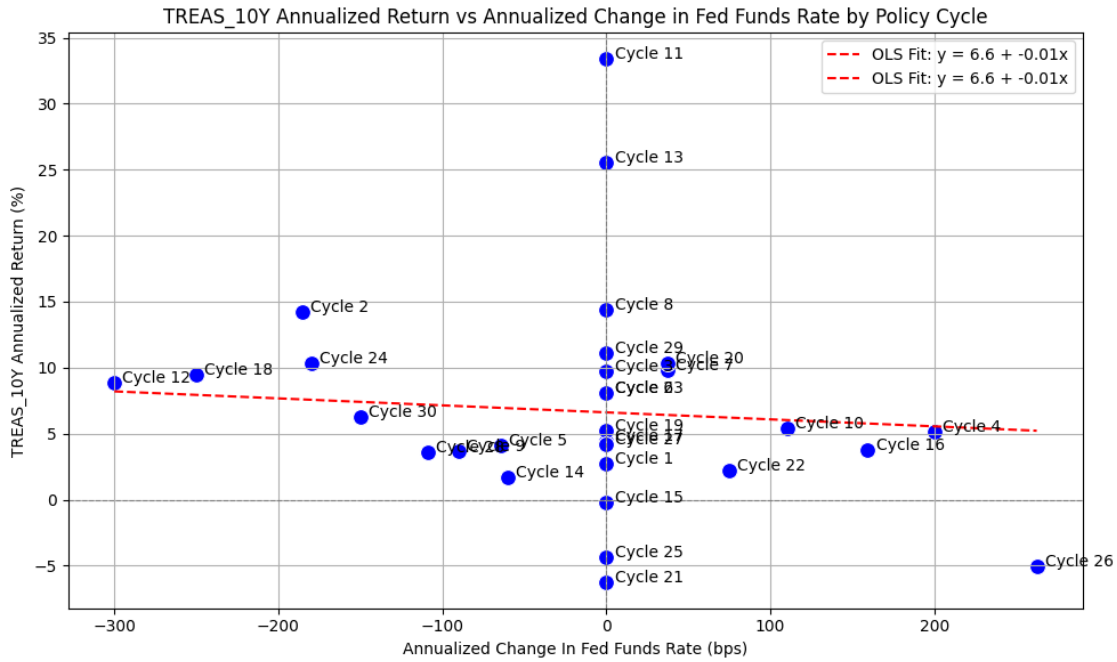
```
                            OLS Regression Results
```

```
==============================================================================
Dep. Variable:     AnnualizedReturnPct   R-squared:                       0.004
Model:                             OLS   Adj. R-squared:                 -0.031
Method:                  Least Squares   F-statistic:                    0.1171
Date:                 Tue, 24 Feb 2026   Prob (F-statistic):              0.735
Time:                         13:51:29   Log-Likelihood:                -110.57
No. Observations:                   30   AIC:                             225.1
Df Residuals:                       28   BIC:                             227.9
Df Model:                            1
Covariance Type:             nonrobust
==============================================================================
============
                               coef    std err          t      P>|t|      [0.025
0.975]
------------------------------------------------------------------------------
------------
const                        6.6113      1.842      3.590      0.001       2.839
10.384
FFR_AnnualizedChange_bps    -0.0053      0.015     -0.342      0.735      -0.037
0.026
==============================================================================
Omnibus:                        9.087   Durbin-Watson:                   1.936
Prob(Omnibus):                  0.011   Jarque-Bera (JB):                9.047
Skew:                          -0.790   Prob(JB):                       0.0109
Kurtosis:                       5.178   Cond. No.                         120.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
Intercept: 6.611340605920211, Slope: -0.005299161729553994
```

And then plot the regression line along with the values:

[31]:
```
plot_scatter_regression_ffr_vs_returns(
    cycle_df=treas_10y_cycles,
    asset_label="TREAS_10Y",
    x_vals=X_vals,
    y_vals=Y_vals,
    intercept=model.params[0],
    slope=model.params[1],
)
```




### 0.6.4 Gold

Finally, we'll run a similar process for gold using XAU_Gold USD Spot.

First, we pull data with the following:

[32]:
```
bb_clean_data(
    base_directory=DATA_DIR,
    fund_ticker_name="XAU_Gold USD Spot",
    source="Bloomberg",
    asset_class="Commodities",
    excel_export=True,
    pickle_export=True,
    output_confirmation=False,
)

gold = load_data(
    base_directory=DATA_DIR,
    ticker="XAU_Gold USD Spot_Clean",
    source="Bloomberg",
    asset_class="Commodities",
    timeframe="Daily",
    file_format="pickle",
)

# Filter GOLD to date range
```

```
gold = gold[(gold.index >= pd.to_datetime(start_date)) & (gold.index <= pd.
 ↪to_datetime(end_date))]

# Drop everything except the "close" column
gold = gold[["Close"]]

# Resample to monthly frequency
gold_monthly = gold.resample("M").last()
gold_monthly["Monthly_Return"] = gold_monthly["Close"].pct_change()
gold_monthly = gold_monthly.dropna()

display(gold_monthly)
```

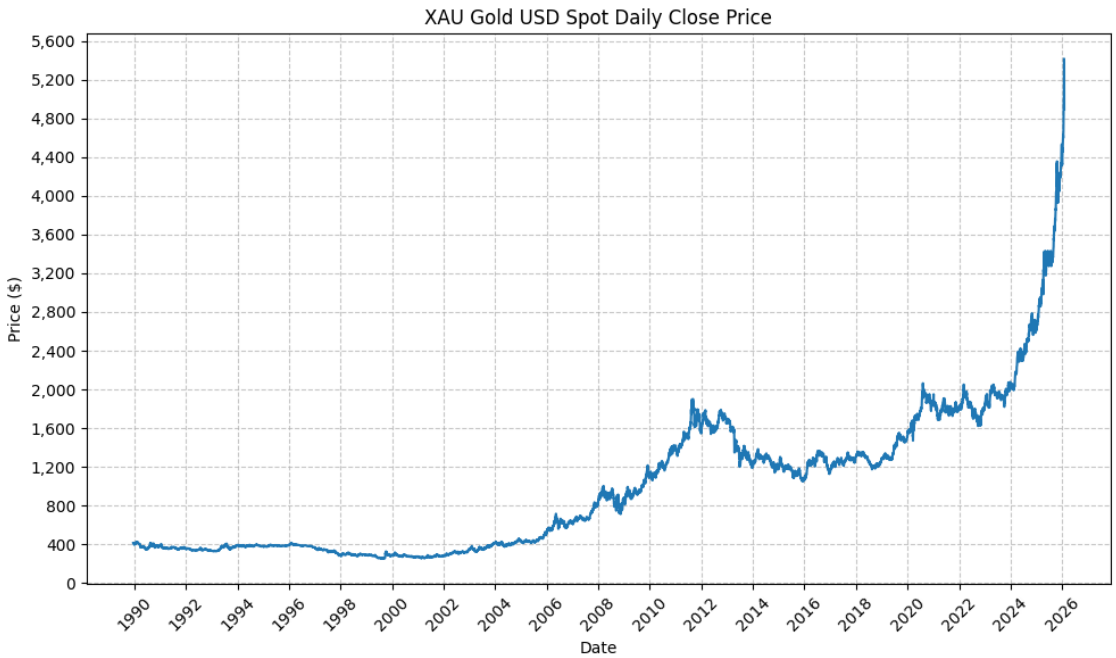
```
                Close  Monthly_Return
Date
1990-01-31  415.05000         0.03439
1990-02-28  407.70000        -0.01771
1990-03-31  368.50000        -0.09615
1990-04-30  367.75000        -0.00204
1990-05-31  363.05000        -0.01278
…                …             …
2025-09-30 3858.96000         0.11920
2025-10-31 4002.92000         0.03731
2025-11-30 4239.43000         0.05908
2025-12-31 4319.37000         0.01886
2026-01-31 4894.23000         0.13309

[433 rows x 2 columns]
```

Next, we can plot the price history before calculating the cycle performance:

[33]:
```
plot_timeseries(
    price_df=gold,
    plot_start_date=None,
    plot_end_date=None,
    plot_columns=["Close"],
    title="XAU Gold USD Spot Daily Close Price",
    x_label="Date",
    x_format="Year",
    x_tick_spacing=2,
    x_tick_rotation=45,
    y_label="Price ($)",
    y_format="Decimal",
    y_format_decimal_places=0,
    y_tick_spacing=400,
    grid=True,
    legend=False,
    export_plot=False,
```

```
    plot_file_name=None,
)
```




Next, we will calculate the performance of gold based on the pre-defined Fed cycles:

[34]:
```
gold_cycles = calc_fed_cycle_asset_performance(
    start_date=cycle_ranges["start_date"],
    end_date=cycle_ranges["end_date"],
    label=cycle_ranges["cycle_label"],
    fed_funds_change=cycle_ranges["fed_funds_change"],
    monthly_returns=gold_monthly,
)

display(gold_cycles)
```

```
       Cycle       Start         End  Months  CumulativeReturn  \
0    Cycle 1  1989-12-31  1990-06-30       6          -0.12224
1    Cycle 2  1990-06-30  1993-03-31      34          -0.06624
2    Cycle 3  1993-03-31  1994-01-31      11           0.15903
3    Cycle 4  1994-01-31  1995-06-30      18          -0.01561
4    Cycle 5  1995-06-30  1996-07-31      14           0.00716
5    Cycle 6  1996-07-31  1997-02-28       8          -0.04468
6    Cycle 7  1997-02-28  1997-09-30       8          -0.02875
7    Cycle 8  1997-09-30  1998-08-31      12          -0.14993
8    Cycle 9  1998-08-31  1999-05-31      10          -0.05621
9   Cycle 10  1999-05-31  2000-11-30      19          -0.05619
```

| | | | | | |
|---|---|---|---|---|---|
| 10 | Cycle 11 | 2000-11-30 | 2000-12-31 | 2 | 0.02678 |
| 11 | Cycle 12 | 2000-12-31 | 2002-06-30 | 19 | 0.16269 |
| 12 | Cycle 13 | 2002-06-30 | 2002-10-31 | 5 | -0.02695 |
| 13 | Cycle 14 | 2002-10-31 | 2003-12-31 | 15 | 0.28404 |
| 14 | Cycle 15 | 2003-12-31 | 2004-05-31 | 6 | -0.00653 |
| 15 | Cycle 16 | 2004-05-31 | 2006-12-31 | 32 | 0.64628 |
| 16 | Cycle 17 | 2006-12-31 | 2007-08-31 | 9 | 0.03904 |
| 17 | Cycle 18 | 2007-08-31 | 2009-07-31 | 24 | 0.43610 |
| 18 | Cycle 19 | 2009-07-31 | 2015-11-30 | 77 | 0.14924 |
| 19 | Cycle 20 | 2015-11-30 | 2016-06-30 | 8 | 0.15742 |
| 20 | Cycle 21 | 2016-06-30 | 2016-11-30 | 6 | -0.03466 |
| 21 | Cycle 22 | 2016-11-30 | 2019-06-30 | 32 | 0.10362 |
| 22 | Cycle 23 | 2019-06-30 | 2019-07-31 | 2 | 0.08288 |
| 23 | Cycle 24 | 2019-07-31 | 2020-09-30 | 15 | 0.33789 |
| 24 | Cycle 25 | 2020-09-30 | 2022-02-28 | 18 | -0.02989 |
| 25 | Cycle 26 | 2022-02-28 | 2024-01-31 | 24 | 0.13485 |
| 26 | Cycle 27 | 2024-01-31 | 2024-08-31 | 8 | 0.21348 |
| 27 | Cycle 28 | 2024-08-31 | 2025-06-30 | 11 | 0.34954 |
| 28 | Cycle 29 | 2025-06-30 | 2025-08-31 | 3 | 0.04825 |
| 29 | Cycle 30 | 2025-08-31 | 2026-01-31 | 6 | 0.48764 |

| | CumulativeReturnPct | AverageMonthlyReturn | AverageMonthlyReturnPct | \ |
|---|---|---|---|---|
| 0 | -12.22430 | -0.02069 | -2.06945 | |
| 1 | -6.62443 | -0.00158 | -0.15839 | |
| 2 | 15.90288 | 0.01462 | 1.46238 | |
| 3 | -1.56130 | -0.00070 | -0.06966 | |
| 4 | 0.71559 | 0.00066 | 0.06552 | |
| 5 | -4.46839 | -0.00519 | -0.51936 | |
| 6 | -2.87498 | -0.00313 | -0.31292 | |
| 7 | -14.99306 | -0.01285 | -1.28489 | |
| 8 | -5.62053 | -0.00517 | -0.51718 | |
| 9 | -5.61857 | -0.00193 | -0.19269 | |
| 10 | 2.67773 | 0.01332 | 1.33221 | |
| 11 | 16.26918 | 0.00844 | 0.84406 | |
| 12 | -2.69484 | -0.00496 | -0.49607 | |
| 13 | 28.40365 | 0.01772 | 1.77192 | |
| 14 | -0.65302 | 0.00044 | 0.04419 | |
| 15 | 64.62831 | 0.01653 | 1.65326 | |
| 16 | 3.90432 | 0.00447 | 0.44660 | |
| 17 | 43.60981 | 0.01766 | 1.76641 | |
| 18 | 14.92391 | 0.00317 | 0.31742 | |
| 19 | 15.74192 | 0.02027 | 2.02679 | |
| 20 | -3.46575 | -0.00449 | -0.44887 | |
| 21 | 10.36165 | 0.00355 | 0.35519 | |
| 22 | 8.28750 | 0.04132 | 4.13180 | |
| 23 | 33.78880 | 0.02043 | 2.04336 | |
| 24 | -2.98862 | -0.00076 | -0.07561 | |
| 25 | 13.48509 | 0.00605 | 0.60516 | |

| | | | |
|---|---|---|---|
| 26 | 21.34824 | 0.02494 | 2.49353 |
| 27 | 34.95424 | 0.02824 | 2.82419 |
| 28 | 4.82481 | 0.01609 | 1.60850 |
| 29 | 48.76396 | 0.06926 | 6.92618 |

| | AnnualizedReturn | AnnualizedReturnPct | Volatility | FedFundsChange | \ |
|---|---|---|---|---|---|
| 0 | -0.22954 | -22.95426 | 0.14885 | 0.00000 | |
| 1 | -0.02390 | -2.39005 | 0.10292 | -0.05250 | |
| 2 | 0.17468 | 17.46838 | 0.17075 | 0.00000 | |
| 3 | -0.01044 | -1.04359 | 0.06717 | 0.03000 | |
| 4 | 0.00613 | 0.61305 | 0.06186 | -0.00750 | |
| 5 | -0.06627 | -6.62715 | 0.11737 | 0.00000 | |
| 6 | -0.04281 | -4.28133 | 0.11886 | 0.00250 | |
| 7 | -0.14993 | -14.99306 | 0.12433 | 0.00000 | |
| 8 | -0.06706 | -6.70614 | 0.12709 | -0.00750 | |
| 9 | -0.03586 | -3.58627 | 0.17333 | 0.01750 | |
| 10 | 0.17181 | 17.18109 | 0.03266 | 0.00000 | |
| 11 | 0.09988 | 9.98819 | 0.11022 | -0.04750 | |
| 12 | -0.06346 | -6.34605 | 0.12104 | 0.00000 | |
| 13 | 0.22141 | 22.14112 | 0.15439 | -0.00750 | |
| 14 | -0.01302 | -1.30178 | 0.20868 | 0.00000 | |
| 15 | 0.20556 | 20.55610 | 0.14568 | 0.04250 | |
| 16 | 0.05239 | 5.23935 | 0.07374 | 0.00000 | |
| 17 | 0.19837 | 19.83731 | 0.24658 | -0.05000 | |
| 18 | 0.02191 | 2.19146 | 0.18253 | 0.00000 | |
| 19 | 0.24519 | 24.51911 | 0.22531 | 0.00250 | |
| 20 | -0.06811 | -6.81139 | 0.19940 | 0.00000 | |
| 21 | 0.03766 | 3.76642 | 0.10758 | 0.02000 | |
| 22 | 0.61239 | 61.23899 | 0.18771 | 0.00000 | |
| 23 | 0.26222 | 26.22222 | 0.14899 | -0.02250 | |
| 24 | -0.02002 | -2.00247 | 0.15388 | 0.00000 | |
| 25 | 0.06529 | 6.52938 | 0.13996 | 0.05250 | |
| 26 | 0.33675 | 33.67502 | 0.11399 | 0.00000 | |
| 27 | 0.38683 | 38.68250 | 0.12923 | -0.01000 | |
| 28 | 0.20741 | 20.74143 | 0.09689 | 0.00000 | |
| 29 | 1.21307 | 121.30714 | 0.16013 | -0.00750 | |

| | FedFundsChange_bps | FFR_AnnualizedChange | FFR_AnnualizedChange_bps | \ |
|---|---|---|---|---|
| 0 | 0.00000 | 0.00000 | 0.00000 | |
| 1 | -525.00000 | -0.01853 | -185.29412 | |
| 2 | 0.00000 | 0.00000 | 0.00000 | |
| 3 | 300.00000 | 0.02000 | 200.00000 | |
| 4 | -75.00000 | -0.00643 | -64.28571 | |
| 5 | 0.00000 | 0.00000 | 0.00000 | |
| 6 | 25.00000 | 0.00375 | 37.50000 | |
| 7 | 0.00000 | 0.00000 | 0.00000 | |
| 8 | -75.00000 | -0.00900 | -90.00000 | |
| 9 | 175.00000 | 0.01105 | 110.52632 | |

| | | | |
|---|---|---|---|
| 10 | 0.00000 | 0.00000 | 0.00000 |
| 11 | -475.00000 | -0.03000 | -300.00000 |
| 12 | 0.00000 | 0.00000 | 0.00000 |
| 13 | -75.00000 | -0.00600 | -60.00000 |
| 14 | 0.00000 | 0.00000 | 0.00000 |
| 15 | 425.00000 | 0.01594 | 159.37500 |
| 16 | 0.00000 | 0.00000 | 0.00000 |
| 17 | -500.00000 | -0.02500 | -250.00000 |
| 18 | 0.00000 | 0.00000 | 0.00000 |
| 19 | 25.00000 | 0.00375 | 37.50000 |
| 20 | 0.00000 | 0.00000 | 0.00000 |
| 21 | 200.00000 | 0.00750 | 75.00000 |
| 22 | 0.00000 | 0.00000 | 0.00000 |
| 23 | -225.00000 | -0.01800 | -180.00000 |
| 24 | 0.00000 | 0.00000 | 0.00000 |
| 25 | 525.00000 | 0.02625 | 262.50000 |
| 26 | 0.00000 | 0.00000 | 0.00000 |
| 27 | -100.00000 | -0.01091 | -109.09091 |
| 28 | 0.00000 | 0.00000 | 0.00000 |
| 29 | -75.00000 | -0.01500 | -150.00000 |

| | Label |
|---|---|
| 0 | Cycle 1, 1989-12-31 to 1990-06-30 |
| 1 | Cycle 2, 1990-06-30 to 1993-03-31 |
| 2 | Cycle 3, 1993-03-31 to 1994-01-31 |
| 3 | Cycle 4, 1994-01-31 to 1995-06-30 |
| 4 | Cycle 5, 1995-06-30 to 1996-07-31 |
| 5 | Cycle 6, 1996-07-31 to 1997-02-28 |
| 6 | Cycle 7, 1997-02-28 to 1997-09-30 |
| 7 | Cycle 8, 1997-09-30 to 1998-08-31 |
| 8 | Cycle 9, 1998-08-31 to 1999-05-31 |
| 9 | Cycle 10, 1999-05-31 to 2000-11-30 |
| 10 | Cycle 11, 2000-11-30 to 2000-12-31 |
| 11 | Cycle 12, 2000-12-31 to 2002-06-30 |
| 12 | Cycle 13, 2002-06-30 to 2002-10-31 |
| 13 | Cycle 14, 2002-10-31 to 2003-12-31 |
| 14 | Cycle 15, 2003-12-31 to 2004-05-31 |
| 15 | Cycle 16, 2004-05-31 to 2006-12-31 |
| 16 | Cycle 17, 2006-12-31 to 2007-08-31 |
| 17 | Cycle 18, 2007-08-31 to 2009-07-31 |
| 18 | Cycle 19, 2009-07-31 to 2015-11-30 |
| 19 | Cycle 20, 2015-11-30 to 2016-06-30 |
| 20 | Cycle 21, 2016-06-30 to 2016-11-30 |
| 21 | Cycle 22, 2016-11-30 to 2019-06-30 |
| 22 | Cycle 23, 2019-06-30 to 2019-07-31 |
| 23 | Cycle 24, 2019-07-31 to 2020-09-30 |
| 24 | Cycle 25, 2020-09-30 to 2022-02-28 |
| 25 | Cycle 26, 2022-02-28 to 2024-01-31 |

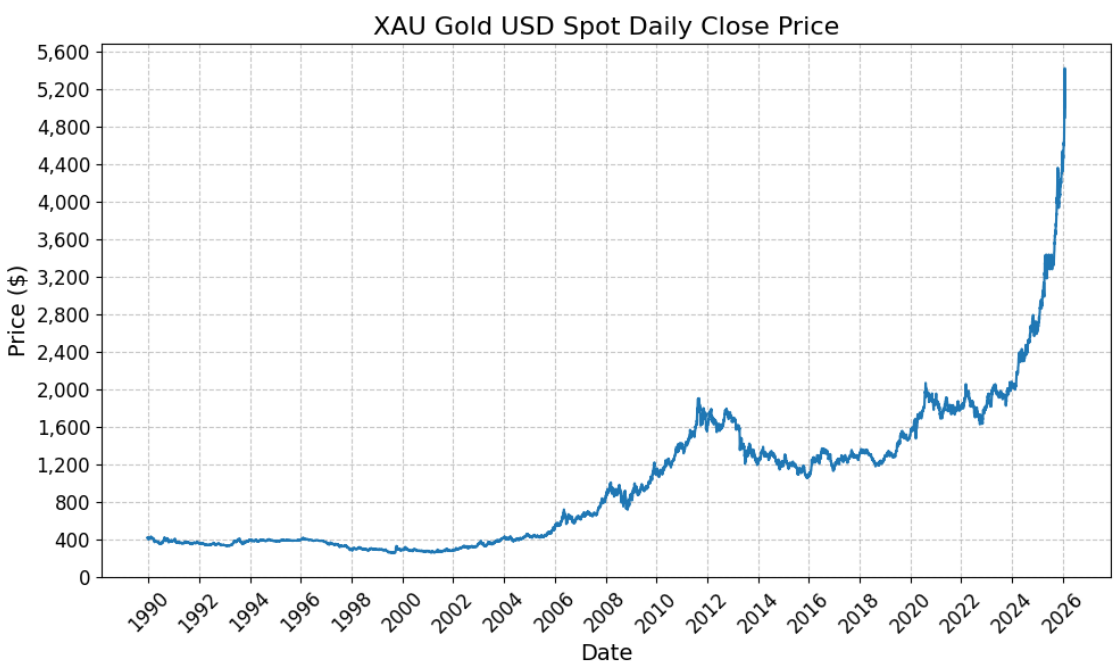
```
26  Cycle 27, 2024-01-31 to 2024-08-31
27  Cycle 28, 2024-08-31 to 2025-06-30
28  Cycle 29, 2025-06-30 to 2025-08-31
29  Cycle 30, 2025-08-31 to 2026-01-31
```

This gives us the following data points:

- Cycle start date
- Cycle end date
- Number of months in the cycle
- Cumulative return during the cycle (decimal and percent)
- Average monthly return during the cycle (decimal and percent)
- Annualized return during the cycle (decimal and percent)
- Return volatility during the cycle
- Cumulative change in FFR during the cycle (decimal and basis points)
- Annualized change in FFR during the cycle (decimal and basis points)

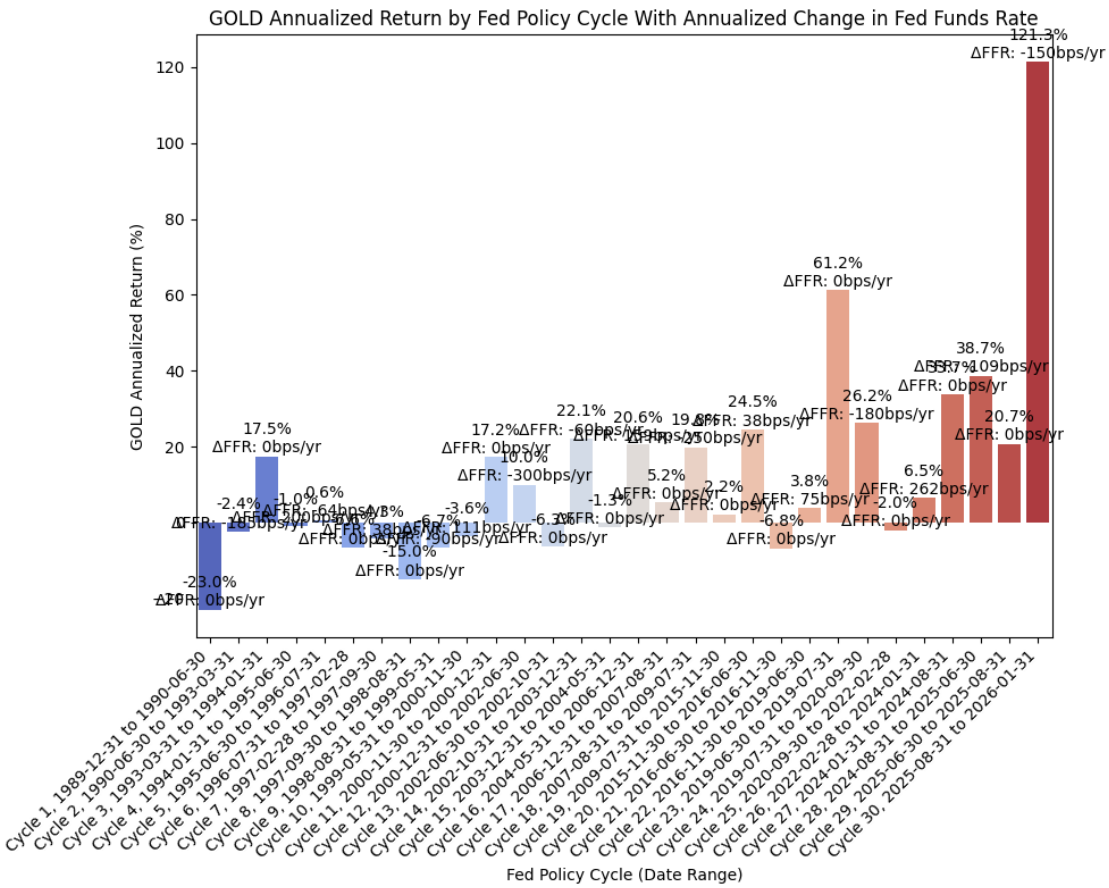
From the above DataFrame, we can then plot the cumulative and annualized returns for each cycle in a bar chart. First, the cumulative returns along with the cumulative change in FFR:

[35]:
```
plot_bar_returns_ffr_change(
    cycle_df=gold_cycles,
    asset_label="GOLD",
    annualized_or_cumulative="Cumulative",
)
```




And then the annualized returns along with the annualized change in FFR:

[36]:
```
plot_bar_returns_ffr_change(
    cycle_df=gold_cycles,
    asset_label="GOLD",
    annualized_or_cumulative="Annualized",
)
```




We see strong returns for gold across several different Fed cycles, so it is difficult to draw any kind of initial conclusion based on the bar charts.

Finally, we can run an OLS regression to check fit:

[37]:
```
df = gold_cycles

#=== Don't modify below this line ===

# Run OLS regression with statsmodels
X = df["FFR_AnnualizedChange_bps"]
y = df["AnnualizedReturnPct"]
X = sm.add_constant(X)
model = sm.OLS(y, X).fit()
```

```
print(model.summary())
print(f"Intercept: {model.params[0]}, Slope: {model.params[1]}")  # Intercept␣
↪and slope

# Calc X and Y values for regression line
X_vals = np.linspace(X.min(), X.max(), 100)
Y_vals = model.params[0] + model.params[1] * X_vals
```

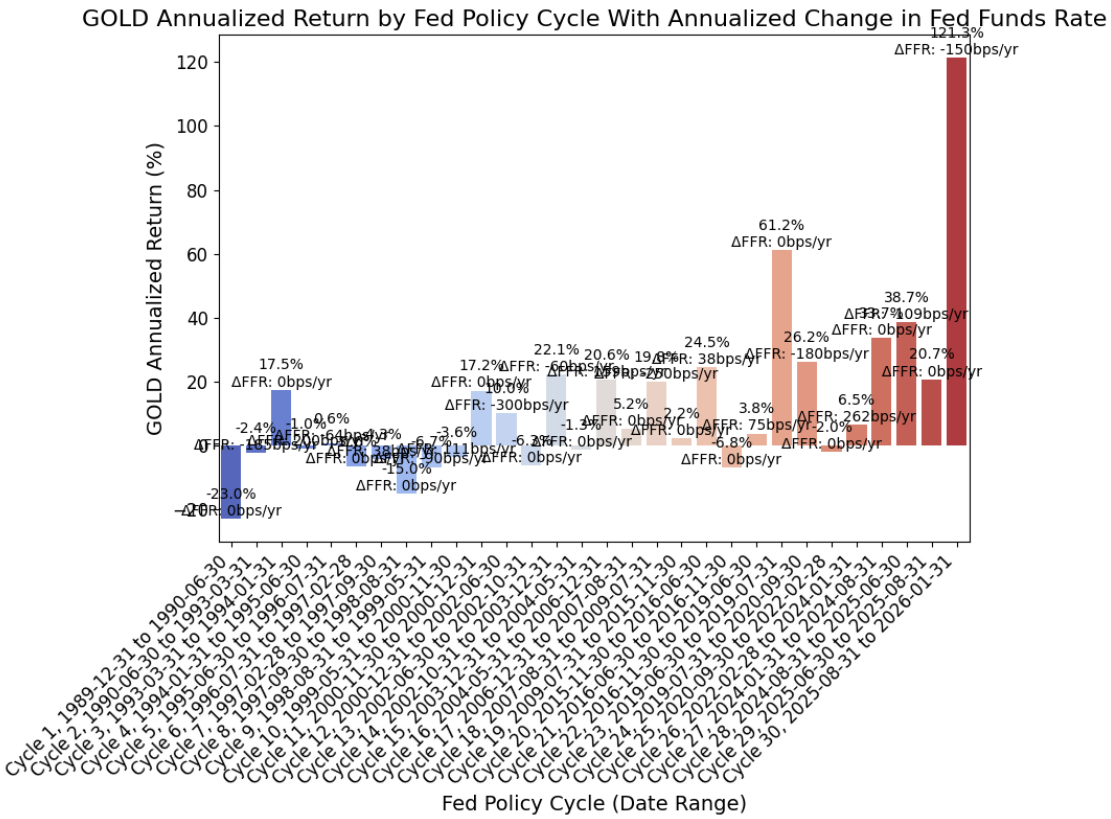
```
                            OLS Regression Results
==============================================================================
Dep. Variable:     AnnualizedReturnPct   R-squared:                       0.064
Model:                             OLS   Adj. R-squared:                  0.030
Method:                  Least Squares   F-statistic:                     1.900
Date:                 Tue, 24 Feb 2026   Prob (F-statistic):              0.179
Time:                         13:51:32   Log-Likelihood:                -140.03
No. Observations:                   30   AIC:                             284.1
Df Residuals:                       28   BIC:                             286.9
Df Model:                            1
Covariance Type:             nonrobust
================================================================================
============
                               coef    std err          t      P>|t|      [0.025
0.975]
--------------------------------------------------------------------------------
------------
const                       11.4670      4.917      2.332      0.027       1.396
21.539
FFR_AnnualizedChange_bps    -0.0570      0.041     -1.378      0.179      -0.142
0.028
==============================================================================
Omnibus:                       28.991   Durbin-Watson:                   1.081
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               62.894
Skew:                           2.093   Prob(JB):                     2.20e-14
Kurtosis:                       8.727   Cond. No.                         120.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
Intercept: 11.467013480569243, Slope: -0.056974234027166226
```

And then plot the regression line along with the values:

[38]:
```
plot_scatter_regression_ffr_vs_returns(
    cycle_df=gold_cycles,
    asset_label="GOLD",
    x_vals=X_vals,
    y_vals=Y_vals,
```

```
    intercept=model.params[0],
    slope=model.params[1],
)
```




It's difficult to draw any strong conclusions with the above plot. Gold has traditionally been considered a hedge for inflation, and while one of the Fed's mandates is to manage inflation, there may not be a conclusion to draw in relationship to the historical returns that gold has exhibited.

## 0.7 Hybrid Portfolio

With the above analysis (somewhat) complete, let's look at the optimal allocation for a portfolio based on the data and the hypothetical historical results.

### 0.7.1 Asset Allocation

We have to be careful with our criteria for when to hold stocks, bonds, or gold, as hindsight bias is certainly possible. So, without overanalyzing the results, let's assume that we hold stocks as the default position during tightening cycles, and then hold bonds during easing cycles when the Fed starts cutting rates, and then resume holding stocks when the Fed stops cutting rates. If there is not any change in FFR, then we still hold stocks.

We can then combine the return series based on the above with the following:

[39]:
```
# Shift the "cycle_filled" column down by one row to create a new column called␣
↪"cycle_invested" that represents the cycle label that an investor would be␣
↪invested in for each month (i.e. the cycle label from the previous month)

fedfunds_grouped_cycles['cycle_invested'] =␣
↪fedfunds_grouped_cycles['cycle_filled'].shift(1)

display(fedfunds_grouped_cycles)
```

```
    start_date  fed_funds_start   end_date  fed_funds_end  fed_funds_change  \
0   1989-12-31          0.08250 1990-01-31        0.08250           0.00000
1   1990-01-31          0.08250 1990-02-28        0.08250           0.00000
2   1990-02-28          0.08250 1990-03-31        0.08250           0.00000
3   1990-03-31          0.08250 1990-04-30        0.08250           0.00000
4   1990-04-30          0.08250 1990-05-31        0.08250           0.00000
..         …                  …        …             …                 …
429 2025-09-30          0.04250 2025-10-31        0.04000          -0.00250
430 2025-10-31          0.04000 2025-11-30        0.04000           0.00000
431 2025-11-30          0.04000 2025-12-31        0.03750          -0.00250
432 2025-12-31          0.03750 2026-01-31        0.03750           0.00000
433 2026-01-31          0.03750        NaT            NaN               NaN

       cycle         cycle_filled  group       cycle_invested
0    Neutral  Modified Tightening      1                 None
1    Neutral  Modified Tightening      1  Modified Tightening
2    Neutral  Modified Tightening      1  Modified Tightening
3    Neutral  Modified Tightening      1  Modified Tightening
4    Neutral  Modified Tightening      1  Modified Tightening
..       …                   …    …                   …
429   Easing               Easing     30               Easing
430  Neutral               Easing     30               Easing
431   Easing               Easing     30               Easing
432  Neutral               Easing     30               Easing
433  Neutral               Easing     30               Easing

[434 rows x 9 columns]
```

[40]:
```
# Reset index to merge on date
stocks_merged = pd.merge_asof(
    spxt_monthly.reset_index(),
    fedfunds_grouped_cycles[['start_date', 'end_date', 'cycle_invested',␣
↪'group']],
    left_on='Date',
    right_on='start_date',
    direction='backward'
)
```

```
# Drop rows where the date falls outside the cycle's end_date
stocks_merged = stocks_merged[stocks_merged['Date'] <=␣
↪stocks_merged['end_date']]

display(stocks_merged)
```

```
          Date       Close  Monthly_Return start_date   end_date  \
0   1990-01-31   353.94000        -0.06713 1990-01-31 1990-02-28
1   1990-02-28   358.50000         0.01288 1990-02-28 1990-03-31
2   1990-03-31         368         0.02650 1990-03-31 1990-04-30
3   1990-04-30   358.81000        -0.02497 1990-04-30 1990-05-31
4   1990-05-31   393.80000         0.09752 1990-05-31 1990-06-30
..         …          …              …         …         …
427 2025-08-31 14304.68000         0.02027 2025-08-31 2025-09-30
428 2025-09-30 14826.80000         0.03650 2025-09-30 2025-10-31
429 2025-10-31 15173.95000         0.02341 2025-10-31 2025-11-30
430 2025-11-30 15211.14000         0.00245 2025-11-30 2025-12-31
431 2025-12-31 15220.45000         0.00061 2025-12-31 2026-01-31

          cycle_invested  group
0    Modified Tightening      1
1    Modified Tightening      1
2    Modified Tightening      1
3    Modified Tightening      1
4    Modified Tightening      1
..                   …    …
427  Modified Tightening     30
428               Easing     30
429               Easing     30
430               Easing     30
431               Easing     30

[432 rows x 7 columns]
```

[41]:
```
# Reset index to merge on date
bonds_merged = pd.merge_asof(
    treas_10y_monthly.reset_index(),
    fedfunds_grouped_cycles[['start_date', 'end_date', 'cycle_invested',␣
↪'group']],
    left_on='Date',
    right_on='start_date',
    direction='backward'
)

# Drop rows where the date falls outside the cycle's end_date
bonds_merged = bonds_merged[bonds_merged['Date'] <= bonds_merged['end_date']]
```

```
display(bonds_merged)
```

```
          Date     Close  Monthly_Return start_date   end_date  \
0   1990-01-31  98.01300        -0.01987 1990-01-31 1990-02-28
1   1990-02-28  97.99000        -0.00023 1990-02-28 1990-03-31
2   1990-03-31  97.98900        -0.00001 1990-03-31 1990-04-30
3   1990-04-30  96.60600        -0.01411 1990-04-30 1990-05-31
4   1990-05-31  99.64700         0.03148 1990-05-31 1990-06-30
..         …       …              …        …         …
427 2025-08-31 641.25800         0.01668 2025-08-31 2025-09-30
428 2025-09-30 645.58400         0.00675 2025-09-30 2025-10-31
429 2025-10-31 650.00500         0.00685 2025-10-31 2025-11-30
430 2025-11-30 656.63600         0.01020 2025-11-30 2025-12-31
431 2025-12-31 652.28800        -0.00662 2025-12-31 2026-01-31

          cycle_invested  group
0    Modified Tightening      1
1    Modified Tightening      1
2    Modified Tightening      1
3    Modified Tightening      1
4    Modified Tightening      1
..                    …    …
427  Modified Tightening     30
428               Easing     30
429               Easing     30
430               Easing     30
431               Easing     30

[432 rows x 7 columns]
```

[42]:
```
# Select the appropriate return based on cycle
stocks_merged['strategy_return'] = stocks_merged.apply(
    lambda row: row['Monthly_Return'] if row['cycle_invested'] in␣
 ↪['Tightening', 'Modified Tightening'] else None,
    axis=1
)

bonds_merged['strategy_return'] = bonds_merged.apply(
    lambda row: row['Monthly_Return'] if row['cycle_invested'] == 'Easing' else␣
 ↪None,
    axis=1
)

# Combine
strategy = pd.concat([stocks_merged, bonds_merged]).
 ↪dropna(subset=['strategy_return'])
strategy = strategy.sort_values('Date')
```

```
display(strategy.head(20))
```

```
         Date     Close  Monthly_Return start_date   end_date  \
0  1990-01-31 353.94000        -0.06713 1990-01-31 1990-02-28
1  1990-02-28 358.50000         0.01288 1990-02-28 1990-03-31
2  1990-03-31       368         0.02650 1990-03-31 1990-04-30
3  1990-04-30 358.81000        -0.02497 1990-04-30 1990-05-31
4  1990-05-31 393.80000         0.09752 1990-05-31 1990-06-30
5  1990-06-30 391.14000        -0.00675 1990-06-30 1990-07-31
6  1990-07-31 102.84500         0.01463 1990-07-31 1990-08-31
7  1990-08-31 100.78200        -0.02006 1990-08-31 1990-09-30
8  1990-09-30 101.74000         0.00951 1990-09-30 1990-10-31
9  1990-10-31 103.65200         0.01879 1990-10-31 1990-11-30
10 1990-11-30 106.43100         0.02681 1990-11-30 1990-12-31
11 1990-12-31 108.24900         0.01708 1990-12-31 1991-01-31
12 1991-01-31 109.52500         0.01179 1991-01-31 1991-02-28
13 1991-02-28 110.10400         0.00529 1991-02-28 1991-03-31
14 1991-03-31 110.43400         0.00300 1991-03-31 1991-04-30
15 1991-04-30 111.59800         0.01054 1991-04-30 1991-05-31
16 1991-05-31 112.07800         0.00430 1991-05-31 1991-06-30
17 1991-06-30 111.50200        -0.00514 1991-06-30 1991-07-31
18 1991-07-31 113.05200         0.01390 1991-07-31 1991-08-31
19 1991-08-31 116.06500         0.02665 1991-08-31 1991-09-30

         cycle_invested  group  strategy_return
0   Modified Tightening      1         -0.06713
1   Modified Tightening      1          0.01288
2   Modified Tightening      1          0.02650
3   Modified Tightening      1         -0.02497
4   Modified Tightening      1          0.09752
5   Modified Tightening      2         -0.00675
6                Easing      2          0.01463
7                Easing      2         -0.02006
8                Easing      2          0.00951
9                Easing      2          0.01879
10               Easing      2          0.02681
11               Easing      2          0.01708
12               Easing      2          0.01179
13               Easing      2          0.00529
14               Easing      2          0.00300
15               Easing      2          0.01054
16               Easing      2          0.00430
17               Easing      2         -0.00514
18               Easing      2          0.01390
19               Easing      2          0.02665
```

[43]:
```
# Calculate cumulative returns and drawdown for spxt
spxt_monthly['Cumulative_Return'] = (1 + spxt_monthly['Monthly_Return']).
 ↪cumprod() - 1
spxt_monthly['Cumulative_Return_Plus_One'] = 1 +␣
 ↪spxt_monthly['Cumulative_Return']
spxt_monthly['Rolling_Max'] = spxt_monthly['Cumulative_Return_Plus_One'].
 ↪cummax()
spxt_monthly['Drawdown'] = spxt_monthly['Cumulative_Return_Plus_One'] /␣
 ↪spxt_monthly['Rolling_Max'] - 1
spxt_monthly.drop(columns=['Cumulative_Return_Plus_One', 'Rolling_Max'],␣
 ↪inplace=True)

# Calculate cumulative returns and drawdown for treas_10y
treas_10y_monthly['Cumulative_Return'] = (1 +␣
 ↪treas_10y_monthly['Monthly_Return']).cumprod() - 1
treas_10y_monthly['Cumulative_Return_Plus_One'] = 1 +␣
 ↪treas_10y_monthly['Cumulative_Return']
treas_10y_monthly['Rolling_Max'] =␣
 ↪treas_10y_monthly['Cumulative_Return_Plus_One'].cummax()
treas_10y_monthly['Drawdown'] = treas_10y_monthly['Cumulative_Return_Plus_One']␣
 ↪/ treas_10y_monthly['Rolling_Max'] - 1
treas_10y_monthly.drop(columns=['Cumulative_Return_Plus_One', 'Rolling_Max'],␣
 ↪inplace=True)

# Convert to DataFrame
portfolio_monthly = strategy[['Date', 'strategy_return']].copy().
 ↪set_index('Date')
portfolio_monthly = portfolio_monthly.rename(columns={'strategy_return':␣
 ↪'Portfolio_Monthly_Return'})

# Calculate cumulative returns and drawdown for the portfolio
portfolio_monthly['Portfolio_Cumulative_Return'] = (1 +␣
 ↪portfolio_monthly['Portfolio_Monthly_Return']).cumprod() - 1
portfolio_monthly['Portfolio_Cumulative_Return_Plus_One'] = 1 +␣
 ↪portfolio_monthly['Portfolio_Cumulative_Return']
portfolio_monthly['Portfolio_Rolling_Max'] =␣
 ↪portfolio_monthly['Portfolio_Cumulative_Return_Plus_One'].cummax()
portfolio_monthly['Portfolio_Drawdown'] =␣
 ↪portfolio_monthly['Portfolio_Cumulative_Return_Plus_One'] /␣
 ↪portfolio_monthly['Portfolio_Rolling_Max'] - 1
portfolio_monthly.drop(columns=['Portfolio_Cumulative_Return_Plus_One',␣
 ↪'Portfolio_Rolling_Max'], inplace=True)

# Merge "spxt_monthly" and "treas_10y_monthly" into "portfolio_monthly" to␣
 ↪compare cumulative returns
portfolio_monthly = portfolio_monthly.join(
```

```
    spxt_monthly['Monthly_Return'].rename('SPXT_Monthly_Return'),
    how='left'
).join(
    spxt_monthly['Cumulative_Return'].rename('SPXT_Cumulative_Return'),
    how='left'
).join(
    spxt_monthly['Drawdown'].rename('SPXT_Drawdown'),
    how='left'
).join(
    treas_10y_monthly['Monthly_Return'].rename('10Y_Monthly_Return'),
    how='left'
).join(
    treas_10y_monthly['Cumulative_Return'].rename('10Y_Cumulative_Return'),
    how='left'
).join(
    treas_10y_monthly['Drawdown'].rename('10Y_Drawdown'),
    how='left'
)
```

[44]:
```
display(portfolio_monthly)
```

```
            Portfolio_Monthly_Return  Portfolio_Cumulative_Return  \
Date
1990-01-31                  -0.06713                    -0.06713
1990-02-28                   0.01288                    -0.05511
1990-03-31                   0.02650                    -0.03007
1990-04-30                  -0.02497                    -0.05429
1990-05-31                   0.09752                     0.03793
…                                …                           …
2025-08-31                   0.02027                    42.28506
2025-09-30                   0.00675                    42.57706
2025-10-31                   0.00685                    42.87548
2025-11-30                   0.01020                    43.32308
2025-12-31                  -0.00662                    43.02959

            Portfolio_Drawdown  SPXT_Monthly_Return  SPXT_Cumulative_Return  \
Date
1990-01-31             0.00000             -0.06713                -0.06713
1990-02-28             0.00000              0.01288                -0.05511
1990-03-31             0.00000              0.02650                -0.03007
1990-04-30            -0.02497             -0.02497                -0.05429
1990-05-31             0.00000              0.09752                 0.03793
…                          …                    …                       …
2025-08-31             0.00000              0.02027                36.70243
2025-09-30             0.00000              0.03650                38.07857
2025-10-31             0.00000              0.02341                38.99354
2025-11-30             0.00000              0.00245                39.09156
2025-12-31            -0.00662              0.00061                39.11610
```

```
            SPXT_Drawdown  10Y_Monthly_Return  10Y_Cumulative_Return  \
Date
1990-01-31        0.00000            -0.01987               -0.01987
1990-02-28        0.00000            -0.00023               -0.02010
1990-03-31        0.00000            -0.00001               -0.02011
1990-04-30       -0.02497            -0.01411               -0.03394
1990-05-31        0.00000             0.03148               -0.00353
…                   …                  …                      …
2025-08-31        0.00000             0.01668                5.41258
2025-09-30        0.00000             0.00675                5.45584
2025-10-31        0.00000             0.00685                5.50005
2025-11-30        0.00000             0.01020                5.56636
2025-12-31        0.00000            -0.00662                5.52288

            10Y_Drawdown
Date
1990-01-31       0.00000
1990-02-28      -0.00023
1990-03-31      -0.00024
1990-04-30      -0.01436
1990-05-31       0.00000
…                   …
2025-08-31      -0.11316
2025-09-30      -0.10717
2025-10-31      -0.10106
2025-11-30      -0.09189
2025-12-31      -0.09790

[432 rows x 9 columns]
```

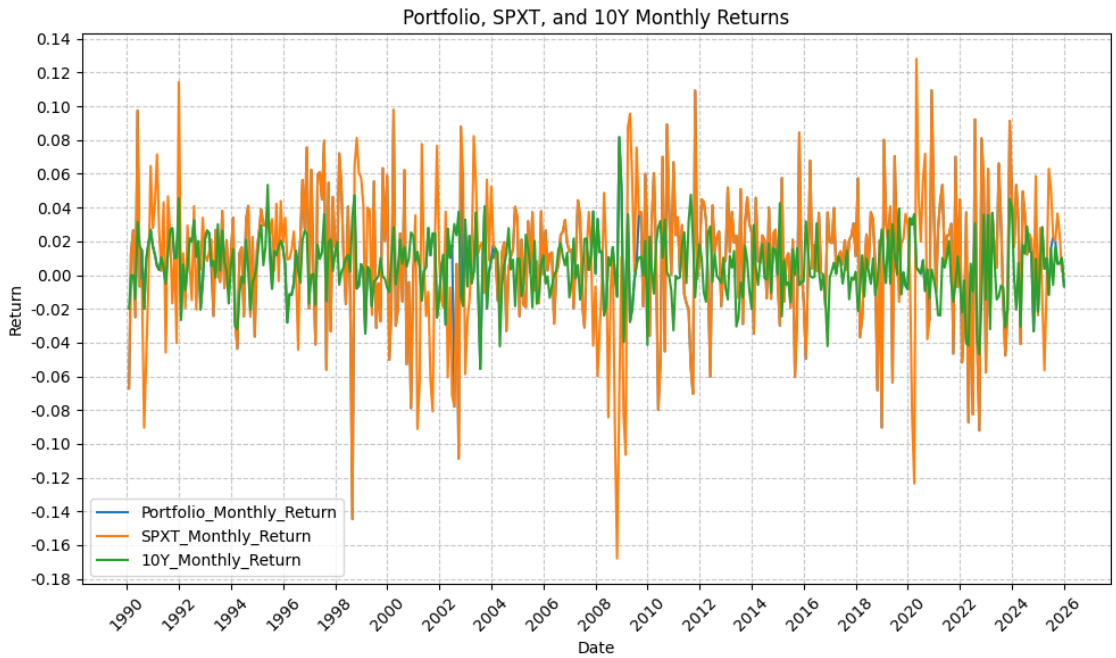
Next, we'll look at performance for the assets and portfolio.

### 0.7.2 Performance Statistics

We can then plot the monthly returns:

[45]:
```
plot_timeseries(
    price_df=portfolio_monthly,
    plot_start_date=start_date,
    plot_end_date=end_date,
    plot_columns=["Portfolio_Monthly_Return", "SPXT_Monthly_Return",␣
 ↪"10Y_Monthly_Return"],
    title="Portfolio, SPXT, and 10Y Monthly Returns",
    x_label="Date",
    x_format="Year",
    x_tick_spacing=2,
    x_tick_rotation=45,
```

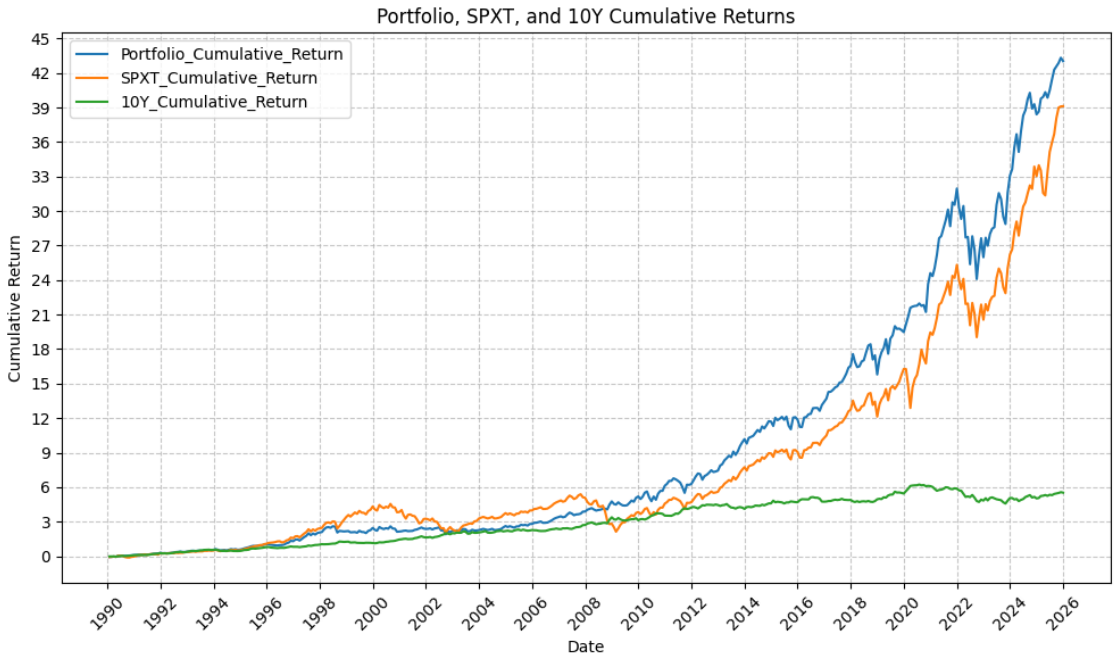
```
    y_label="Return",
    y_format="Decimal",
    y_format_decimal_places=2,
    y_tick_spacing=0.02,
    grid=True,
    legend=True,
    export_plot=False,
    plot_file_name=None,
)
```




And cumulative returns:

[46]:
```
plot_timeseries(
    price_df=portfolio_monthly,
    plot_start_date=start_date,
    plot_end_date=end_date,
    plot_columns=["Portfolio_Cumulative_Return", "SPXT_Cumulative_Return",␣
↪"10Y_Cumulative_Return"],
    title="Portfolio, SPXT, and 10Y Cumulative Returns",
    x_label="Date",
    x_format="Year",
    x_tick_spacing=2,
    x_tick_rotation=45,
    y_label="Cumulative Return",
    y_format="Decimal",
```

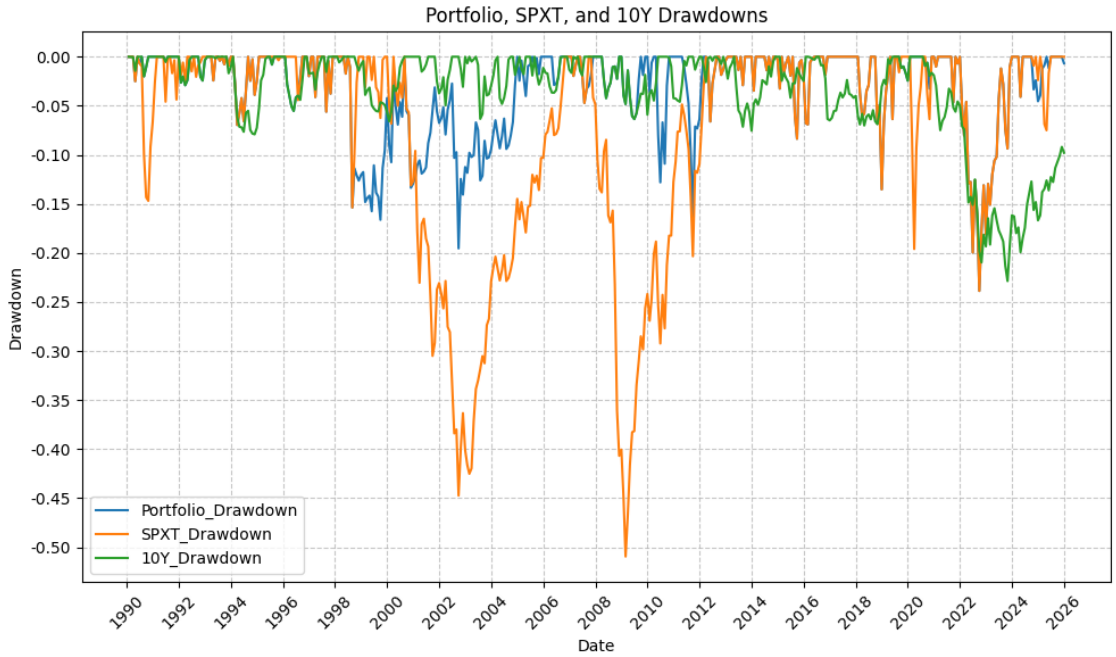
```
    y_format_decimal_places=0,
    y_tick_spacing=3,
    grid=True,
    legend=True,
    export_plot=False,
    plot_file_name=None,
)
```




And drawdowns:

[47]:
```
plot_timeseries(
    price_df=portfolio_monthly,
    plot_start_date=start_date,
    plot_end_date=end_date,
    plot_columns=["Portfolio_Drawdown", "SPXT_Drawdown", "10Y_Drawdown"],
    title="Portfolio, SPXT, and 10Y Drawdowns",
    x_label="Date",
    x_format="Year",
    x_tick_spacing=2,
    x_tick_rotation=45,
    y_label="Drawdown",
    y_format="Decimal",
    y_format_decimal_places=2,
    y_tick_spacing=0.05,
    grid=True,
    legend=True,
```

```
    export_plot=False,
    plot_file_name=None,
)
```




Finally, we can run the stats on the hybrid portfolio, SPY, and TLT with the following code:

[48]:
```
port_sum_stats = summary_stats(
    fund_list=["Portfolio", "SPXT", "10Y"],
    df=portfolio_monthly[["Portfolio_Monthly_Return"]],
    period="Monthly",
    use_calendar_days=False,
    excel_export=False,
    pickle_export=False,
    output_confirmation=False,
)

spy_sum_stats = summary_stats(
    fund_list=["Portfolio", "SPXT", "10Y"],
    df=portfolio_monthly[["SPXT_Monthly_Return"]],
    period="Monthly",
    use_calendar_days=False,
    excel_export=False,
    pickle_export=False,
    output_confirmation=False,
)
```

```
tlt_sum_stats = summary_stats(
    fund_list=["Portfolio", "SPXT", "10Y"],
    df=portfolio_monthly[["10Y_Monthly_Return"]],
    period="Monthly",
    use_calendar_days=False,
    excel_export=False,
    pickle_export=False,
    output_confirmation=False,
)

sum_stats = pd.concat([port_sum_stats, spy_sum_stats, tlt_sum_stats])
sum_stats
```

[48]:

```
                          Annual Mean Return (Arithmetic)  \
Portfolio_Monthly_Return                          0.11267
SPXT_Monthly_Return                               0.11392
10Y_Monthly_Return                                0.05420

                          Annualized Volatility  Annualized Sharpe Ratio  \
Portfolio_Monthly_Return                0.11897                  0.94701
SPXT_Monthly_Return                     0.14735                  0.77311
10Y_Monthly_Return                      0.06331                  0.85606

                          CAGR (Geometric)  Monthly Max Return  \
Portfolio_Monthly_Return           0.11086             0.10946
SPXT_Monthly_Return                0.10799             0.12819
10Y_Monthly_Return                 0.05347             0.08169

                         Monthly Max Return (Date)  Monthly Min Return  \
Portfolio_Monthly_Return                2020-11-30            -0.14458
SPXT_Monthly_Return                     2020-04-30            -0.16795
10Y_Monthly_Return                      2008-11-30            -0.05558

                         Monthly Min Return (Date)  Max Drawdown       Peak  \
Portfolio_Monthly_Return                1998-08-31      -0.23867 2021-12-31
SPXT_Monthly_Return                     2008-10-31      -0.50949 2007-10-31
10Y_Monthly_Return                      2003-07-31      -0.22865 2020-07-31

                            Trough Recovery Date  Days to Recovery  MAR Ratio
Portfolio_Monthly_Return 2022-09-30    2023-12-31         457.00000    0.46449
SPXT_Monthly_Return      2009-02-28    2012-03-31        1127.00000    0.21196
10Y_Monthly_Return       2023-10-31           NaT               NaN    0.23386
```

Based on the above, our hybrid portfolio outperforms both stocks and bonds, with lower drawdowns.

## 0.8 Conclusions

This was a interesting exercise to evaluate the performance of different asset classes during Fed tightening and easing cycles. The results are not particularly surprising, but it is interesting to see the data and plots to confirm the economic intuition that stocks perform well during tightening cycles (economic strength) and bonds perform well during easing cycles (economic weakness). The results are certainly dependent on the specific time period or regime, and also on the assumption made for how to handle the periods of neutral policy (i.e. no change in FFR).

## 0.9 Future Investigation

A couple of ideas sound intriguing for future investigation:

- Does a commodity index (such as GSCI) exhibit differing behavior than gold?
- How does leverage affect the returns that are observed for the hybrid portfolio, stocks, and bonds?
- Do other Fed tightening/loosening cycles exhibit the same behavior for returns?

## 0.10 References


1. https://fred.stlouisfed.org/series/DFEDTARU
2. https://fred.stlouisfed.org/series/DFEDTARL
3. https://fred.stlouisfed.org/series/DFEDTAR